

EmotionWave: GAN-AUGMENTED CUSTOM DATASET USING EEG HEADSET FOR EMOTION RECOGNITION

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Rezumat

Această teză prezintă dezvoltarea setului de date EmotionWave pentru recunoașterea emoțiilor bazată pe EEG folosind headset-ul EMOTIV EPOC X și aplicarea Rețelelor Generativ Adversative (GAN) pentru augmentarea datelor în scopul creșterii dimensiunii și diversității setului de date. Obiectivul principal a fost îmbunătățirea acurateței recunoașterii emoțiilor, abordând în același timp provocările legate de variabilitatea și artefactele datelor EEG. Studiul utilizează informații din domeniul temporal și al frecvenței, folosind arhitecturi avansate de rețele neuronale. Experimentele au demonstrat potențialul încorporării datelor augmentate cu GAN, atingând o acuratețe maximă de evaluare de 66.45% cu modele pre-antrenate fără augmentare GAN, dar perfecționate cu date GAN. În ciuda provocărilor legate de calitatea setului de date și complexitatea antrenării GAN, cercetarea subliniază eficiența GAN în îmbunătățirea performanței modelului. Contribuțiile cheie includ crearea setului de date EmotionWave, implementarea GAN pentru augmentarea datelor și experimente cuprinzătoare. Cercetările viitoare vor viza îmbunătățirea tehnicilor de preprocesare, echilibrarea dinamicii antrenării GAN și explorarea strategiilor avansate de augmentare a datelor pentru a spori robustețea și acuratețea sistemelor de recunoaștere a emoțiilor bazate pe EEG.

Abstract

This thesis presents the development of the EmotionWave dataset for EEG-based emotion recognition using the EMOTIV EPOC X headset and the application of Generative Adversarial Networks (GANs) for data augmentation to enhance dataset size and diversity. The primary objective was to improve emotion recognition accuracy while addressing challenges related to EEG data variability and noise. The study leverages both time and frequency domain information, utilizing advanced neural network architectures. Extensive experimentation demonstrated the potential of incorporating GAN-augmented data, achieving a maximum evaluation accuracy of 66.45% with models pre-trained without GAN augmentation but fine-tuned with GAN data. Despite challenges in dataset quality and GAN training complexities, the research underscores the effectiveness of GANs in enhancing model performance. Key contributions include the creation of the EmotionWave dataset, the implementation of GANs for data augmentation, and comprehensive experimentation. Future work will focus on improving preprocessing techniques, balancing GAN training dynamics, and exploring advanced data augmentation strategies to further enhance the robustness and accuracy of EEG-based emotion recognition systems.

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Chapter 1

Introduction

1.1 Context

Emotion recognition through EEG (Electroencephalography) is an emerging field that combines neuroscience, psychology, and artificial intelligence, with significant potential applications in healthcare, entertainment, and human-computer interaction. EEG measures electrical activity in the brain through non-invasive scalp electrodes, capturing data that reflects cognitive and emotional states. This method provides valuable insights into brain function, making it an important tool for understanding human emotions.

The growing interest in EEG-based emotion recognition is driven by its potential to enhance our interaction with technology. In healthcare, it can result in advanced diagnostic tools for mental health conditions like depression and anxiety, allowing for more personalized treatment plans. In human-computer interaction, it promises to revolutionize how we interact with devices by creating systems that adapt to our emotional states in real-time. For instance, adaptive learning platforms can tailor educational content based on a learner's emotional engagement, while gaming systems can adjust difficulty levels to maintain player engagement.

Recent advancements in machine learning and deep learning have significantly improved the accuracy and robustness of emotion recognition systems. Techniques such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), including Long Short-Term Memory (LSTM) networks, have shown promise in effectively classifying emotional states from EEG signals. Additionally, the integration of multimodal data, such as combining EEG with other physiological signals, has been explored to enhance the reliability of emotion recognition models.

However, several challenges persist in this domain. The variability in EEG signals across different individuals and sessions poses a significant hurdle, often necessitating advanced preprocessing and feature extraction techniques. Moreover, the development of algorithms that generalize well across diverse datasets remains an ongoing research focus. Addressing these challenges is crucial for the deployment of practical and scalable EEG-based emotion recognition systems.

In summary, EEG-based emotion recognition is a promising area of research with the potential to innovate technology and healthcare. This thesis aims to explore advanced methods

to improve the accuracy and applicability of these systems. Through rigorous experimentation and analysis, this work seeks to address existing challenges and contribute to the ongoing development of EEG technology.

1.2 Motivation

The motivation for this research stems from the need to enhance the accuracy and applicability of EEG-based emotion recognition systems. Traditional datasets and models often face limitations due to the inherent variability and noise in EEG data, which can significantly impact the performance of emotion recognition systems. By creating a custom dataset, EmotionWave, and employing advanced neural network architectures, this thesis aims to address these challenges. The use of GANs for data augmentation aims to increase the dataset size and diversity, potentially improving model robustness and generalization. The ultimate goal is to develop an emotion recognition system that can be reliably used in real-world applications, from healthcare to human-computer interaction.

1.3 Objective

The primary objective of this thesis is to develop a robust and accurate EEG-based emotion recognition system. To achieve this, the following specific objectives are set:

- Create a custom EEG dataset, EmotionWave, using the EMOTIV EPOC X headset.
- Augment the dataset using Generative Adversarial Networks (GANs) to increase its size and diversity.
- Adapt and optimize the Joint Contrastive Learning framework with Feature Alignment (JCFA) for the 14-channel configuration of the EPOC X headset.
- Conduct extensive experimentation to fine-tune the model, ensuring it effectively captures the emotional states from the EEG signals.
- Evaluate the model's performance using both the custom dataset and a baseline dataset (SEED) to compare and analyze results.

1.4 Proposed Solution

This work proposes a novel approach to EEG-based emotion recognition by adapting a model inspired by the Joint Contrastive Learning framework with Feature Alignment (JCFA) [26] to the EMOTIV EPOC X headset. The primary goal is to optimize this model for the EPOC X's hardware constraints and fine-tune it to effectively use the 14 EEG channels provided by the headset. Additionally, a Generative Adversarial Network (GAN) was employed to augment the dataset created using the EPOC X, increasing its size and diversity.

The model is designed to extract robust representations from EEG signals by jointly learning from the time and frequency domains through self-supervised learning. This approach employs contrastive learning to align embeddings from these domains into a shared latent space, improving emotion recognition accuracy. This research builds upon this architecture, applying it to work with the EPOC X's 14-channel configuration. Additionally, hyper-parameter tuning experiments, detailed in Section 4.1, were important in fine-tuning the training process.

1.4.1 Dataset Creation and Augmentation

The dataset creation process, detailed in Section 3.2, involved recording EEG data using the EMOTIV EPOC X headset, which provides 14 EEG channels (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4) at a sampling rate of 200 SPS (Samples Per Second). Four separate experiments were conducted, each lasting approximately 55 minutes. The experimental protocol included:

1. **Instructions for the participant**
2. **Headset calibration** (15s with eyes open looking at one spot, 15s with eyes closed)
3. **Each trial consisting of:**
 - **5s hint**
 - **2-4 minute video clip intended to elicit a specific emotion**
 - **Self-assessment by the participant**
 - **15s rest period**

Using the EmotivPRO Builder, the EEG data were automatically marked with events such as the start and stop of each video. The order of the emotions presented was positive, negative, neutral, and then repeating in this sequence.

To enhance the dataset, a Generative Adversarial Network (GAN) was employed to augment the recorded data, increasing the dataset size and diversity. This augmented dataset was used alongside the original recordings to train the model. For comparison, the model was also trained on the SEED dataset, which includes a different set of EEG recordings and labeling conventions.

1.4.2 Pre-training

The pre-training phase, detailed in Section 3.4.2, focused on learning generalizable representations from the EEG data without using labeled information. This phase aimed to initialize the model with features that effectively capture the underlying patterns in EEG signals. The key steps in the pre-training phase included:

1. **Data Augmentation:**

- **Time Domain:** Gaussian noise added to the original time-based signals to generate weakly augmented samples.
- **Frequency Domain:** Original time-based signals transformed into the frequency spectrum using Fourier Transformation, with perturbations introduced by adding or removing frequency components.

2. Contrastive Encoders:

- **Time-based Encoder (ET):** A 2-layer transformer processing time-domain signals.
- **Frequency-based Encoder (EF):** A 2-layer transformer processing frequency-domain signals.

3. Cross-Space Projectors:

- **Time-based Projector (PT):** Maps time-based embeddings to a shared latent space.
- **Frequency-based Projector (PF):** Maps frequency-based embeddings to the same space, ensuring the features extracted from both domains are comparable.

4. Loss Functions:

- **Time Domain Loss (LT):** NT-Xent loss measuring the similarity between augmented and original time-based signals.
- **Frequency Domain Loss (LF):** NT-Xent loss measuring the similarity between augmented and original frequency-based signals.
- **Alignment Loss (LA):** Ensures coherence between time and frequency embeddings within the shared latent space.

1.4.3 Fine-Tuning

During the fine-tuning phase, detailed in Section 3.4.3, the model was trained with labeled EEG data to optimize its performance for emotion recognition. This phase incorporated the spatial relationships between EEG channels, leveraging a Graph Convolutional Network (GCN) to enhance spatial feature extraction.

1. GCN Construction:

- **Adjacency Matrix:** Created an adjacency matrix using cosine similarity between EEG channels to capture spatial features by representing the relationships between different EEG channels.
- **Chebyshev Graph Convolution:** Employed K-order Chebyshev graph convolution to extract these spatial features more effectively.

2. Emotion Classifier:

- A 3-layer fully connected network with hidden layers of dimensions 1024 and 128, followed by an output layer corresponding to the number of emotion classes.

3. Loss Functions:

- **Cross-Space Losses (LT, LF, LA):** NT-Xent losses applied similarly as in the pre-training phase.
- **Classification Loss:** Cross-entropy loss for training the emotion classifier to minimize the error in predicting the correct emotion classes.

By adapting this model to the EMOTIV EPOC X's hardware and focusing on fine-tuning techniques specific to the headset's capabilities, this work aims to contribute to the field of EEG-based emotion recognition. The proposed solution highlights the flexibility of the model framework and underscores the importance of tailoring machine learning models to specific hardware configurations, potentially enhancing the accessibility and effectiveness of emotion recognition technologies in practical applications. Detailed hyper-parameter tuning experiments are discussed in Section 4.1, which further refined the training process for this EEG-based emotion recognition model.

1.5 Theoretical Background

1.5.1 Generative Adversarial Networks (GANs)

In recent years, Generative Adversarial Networks (GANs) [31] have gained significant traction in deep learning. A GAN consists of two main components: the generator (G) and the discriminator (D), which are trained in opposition to each other. The discriminator's role is to differentiate between real and generated (fake) data, while the generator aims to produce fake data that is indistinguishable from real data to the discriminator.

The discriminator (D) is trained to accurately classify input data as real or generated. Conversely, the generator (G) receives a latent noise variable (z) as input and strives to create data that the discriminator cannot easily classify as fake. The training objective involves minimizing the following function for the generator and maximizing it for the discriminator:

$$\min_G \max_D V(G, D) = \mathbb{E}_{x \sim P_{\text{data}}} [\log D(x)] + \mathbb{E}_{z \sim P_z(z)} [\log(1 - D(G(z)))].$$

This objective function ensures that the discriminator becomes increasingly proficient at distinguishing real data from fake data, while the generator continuously enhances its ability to produce realistic data.

Figure 1.1 presents a schematic diagram of the GAN architecture, illustrating the interaction between the generator and discriminator and the adversarial training process.

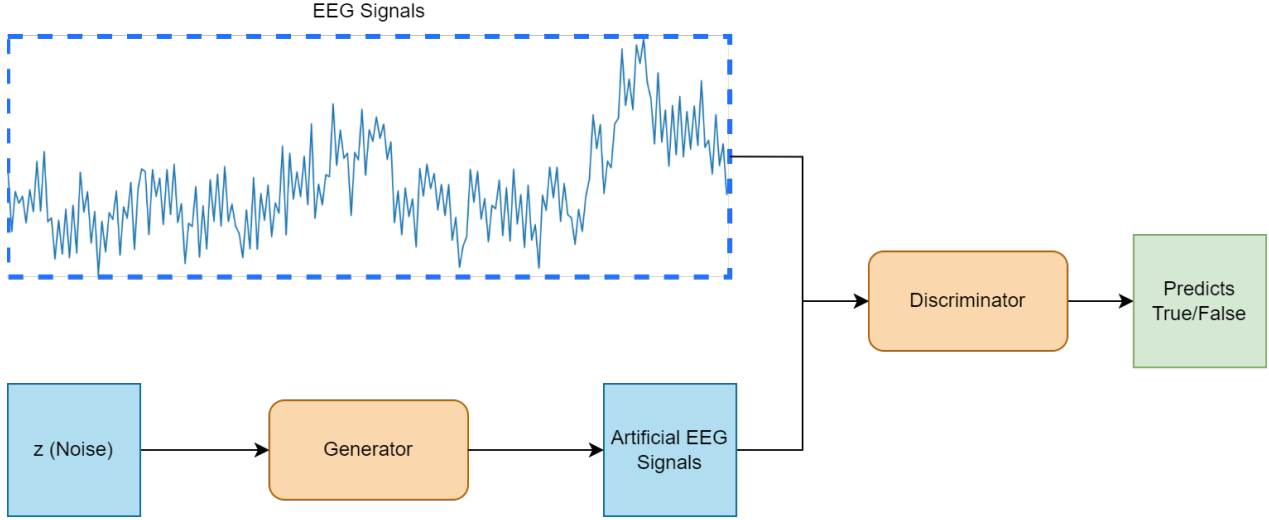


Figure 1.1: GAN architecture for EEG data augmentation.

1.5.2 K-order Chebyshev Graph Convolution

The K-order Chebyshev graph convolution is a technique used in graph signal processing to extend convolutional neural networks (CNNs) to operate on graph-structured data. This method uses Chebyshev polynomials of the graph Laplacian to design filters that are localized and computationally efficient [30].

The graph Laplacian, denoted as L , is an operator in spectral graph theory characterized by its eigenvalues and eigenvectors, which form the graph Fourier basis. Convolution on a graph is defined in the spectral domain using the graph Fourier transform. However, directly applying spectral methods can be computationally expensive due to eigenvalue decomposition and matrix multiplications, which scale quadratically with the number of graph nodes [30].

To overcome these challenges, the K-order Chebyshev polynomial approximation provides a more scalable solution. The convolutional filter is parameterized as a polynomial of the Laplacian:

$$g_{\theta}(\Lambda) = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{\Lambda}),$$

where θ_k are the polynomial coefficients, Λ represents the diagonal matrix of the Laplacian's eigenvalues, and T_k denotes the Chebyshev polynomial of order k evaluated at the scaled Laplacian $\tilde{\Lambda}$. This polynomial expansion confines the filter's support to K neighbors, ensuring localized convolution with a computational complexity of $O(K|E|)$ [30].

The Chebyshev polynomial basis, defined recursively, allows for efficient computation of the filtering operation. Given a signal x on the graph, the filtering operation is performed as:

$$y = g_{\theta}(L)x = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{L})x,$$

where $T_k(\tilde{L})$ can be computed using the recurrence relation $T_k(\tilde{L})x = 2\tilde{L}T_{k-1}(\tilde{L})x - T_{k-2}(\tilde{L})x$

with initial conditions $T_0 = I$ and $T_1 = \tilde{L}$. This recursive formulation reduces computational overhead and leverages the sparsity of the graph Laplacian, making it suitable for large-scale graphs [30].

In practice, the adjacency matrix of the graph is normalized, and the Laplacian is scaled to ensure the eigenvalues lie in the range $[-1, 1]$. The scaled Laplacian $\tilde{L}$ is computed as:

$$\tilde{L} = \frac{2L}{\lambda_{\max}} - I,$$

where $\lambda_{\max}$ is the largest eigenvalue of L . The Chebyshev polynomials $T_k(\tilde{L})$ are then used to approximate the convolutional filter.

In summary, the K-order Chebyshev graph convolution is a practical method for extending CNNs to graph-structured data. It combines spectral graph theory with efficient computation, ensuring that the learned filters are localized and computationally manageable. This approach is particularly effective for dense graphs, as demonstrated in the PyTorch implementation using the **BatchDenseChebConv** class, which efficiently computes the Chebyshev spectral graph convolution for dense adjacency matrices [30].

1.5.3 Normalized Temperature-scaled Cross Entropy Loss (NT-Xent)

Normalized Temperature-scaled Cross Entropy Loss (NT-Xent) is an important objective function in contrastive learning. It aims to learn an embedding space where similar examples are close together and dissimilar examples are far apart. This is useful for tasks such as image retrieval, face verification, and fine-grained object recognition [26, 32, 33].

The NT-Xent loss for a given batch of paired examples $\{(x_i, \tilde{x}_i)\}_{i=1}^N$ and their corresponding embeddings $\{(h_i, \tilde{h}_i)\}_{i=1}^N$ is defined as:

$$\mathcal{L}_{\text{NT-Xent},i} = -\log \frac{\exp(\text{sim}(h_i, \tilde{h}_i)/\tau)}{\sum_{j=1, j \neq i}^N \left[\exp(\text{sim}(h_i, h_j)/\tau) + \exp(\text{sim}(h_i, \tilde{h}_j)/\tau) \right]},$$

where:

- h_i and $\tilde{h}_i$ are the normalized embeddings of the anchor and its positive pair, respectively.
- $\text{sim}(a, b)$ denotes the cosine similarity between embeddings a and b .
- τ is the temperature parameter that controls the sharpness of the similarity distribution.
- The sum is taken over all $j \neq i$, ensuring that the anchor example is not compared with itself.

The numerator of the NT-Xent loss represents the exponential similarity of the anchor-positive pair, scaled by the temperature parameter τ . The denominator includes the exponential similarities of the anchor to all other examples in the batch, excluding itself, plus

the similarities to all positive pairs. This ensures that the similarity to the positive pair is evaluated in the context of all other examples.

The primary goal of the NT-Xent loss is to maximize the similarity between positive pairs while minimizing the similarity with negative pairs, thereby creating a well-structured embedding space where similar examples are close, and dissimilar examples are far apart. The temperature parameter τ helps fine-tune the sensitivity of this similarity measure, with lower values leading to sharper distinctions between pairs.

In summary, NT-Xent loss uses temperature scaling and normalized embeddings to perform contrastive learning effectively, resulting in improved performance in various deep metric learning tasks.

Chapter 2

Related Work

2.1 BCI Methods

2.1.1 Invasive Methods

2.1.1.1 Intracortical Microelectrodes (IM) or Intracortical Neuron Recording (INR)

Intracortical microelectrodes (IM), also known as intracortical neuron recording (INR), have been a significant focus in neuroscience and clinical research. These electrodes are capable of recording rapidly changing action potentials (spikes) from neurons, which are the fundamental units of neural electrical activity. The high-quality recording of these signals is critical for accurate neuronal activity evaluation, with signal quality often assessed through metrics such as the signal-to-noise ratio (SNR), long-term recording stability, and single-unit recording capability [5].

One of the key challenges in the development and application of IMs is achieving long-term stability and biocompatibility. Long-term stability is essential for chronic implantation, where the electrode needs to remain stable in the extracellular fluid environment and resist degradation factors like cracking, delamination, peeling, and corrosion. The choice of materials and processing methods play a crucial role in maintaining this stability. Biocompatibility is another essential requirement, as the electrode materials must not elicit a significant immune response from the surrounding tissue [5].

The tissue response to implanted electrodes can severely affect signal recording quality. This response is categorized into acute and chronic immune responses, primarily driven by mechanical damage from electrode insertion and prolonged exposure. Innovative solutions, such as increasing electrode compliance, soft tethering, and reducing the electrode's physical footprint, have been proposed to mitigate these effects. Additionally, enhancing electrode softness through biocompatible coatings, incorporating anti-inflammatory drugs, bioactive molecules, and certain cells can alleviate inflammatory responses and promote neuronal growth towards the electrodes [5].

In summary, intracortical microelectrodes are pivotal in recording neuronal activity with high fidelity, offering potential in various neuroscientific and clinical applications. However, challenges in long-term stability, biocompatibility, and tissue response management must be

addressed to optimize their functionality and efficacy.

2.1.2 Semi-Invasive Methods

2.1.2.1 Electrocorticography (ECoG)

Electrocorticography (ECoG) is an advanced brain imaging technique employed in brain-computer interface (BCI) research. Unlike non-invasive methods, ECoG involves surgically implanted electrodes to capture local synaptic field potentials of neuronal populations, making it an invasive procedure. This technique offers high spatial and temporal resolution of approximately milliseconds, and excellent signal quality with minimal noise [1, 14].

ECoG's proficiency is not confined to speech BCI but extends to a broader spectrum of applications, including the decoding of complex mental states such as intention decoding from motor areas and speech production [14]. Its capabilities in neural speech decoding include encompassing semantic, auditory, and articulatory representations. Semantic decoding interprets concepts or words in the early stages of speech production across various brain regions. Auditory decoding focuses on acoustic waveforms and spectrotemporal features, requiring recordings from auditory association cortices. Articulatory decoding captures phonation and gestures from the vocal tract and mouth muscles, predominantly in later speech production stages in areas like Broca's area and the ventral sensorimotor cortex (vSMC) [1].

Significantly, ECoG employs somatotopic mapping in the motor cortex, extending its utility to speech articulators. This is exemplified in the vSMC, where ECoG can distinguish phonetic speech segments based on articulatory features, underlining its potential for real-time neural speech decoding. The vSMC's role in speech perception suggests its involvement in envisioning speech intention [1].

Comparative analysis reveals that ECoG offers several advantages over Electroencephalography (EEG). EEG is known for its excellent temporal resolution and ability to cover most of the brain surface. However, it suffers from poor spatial resolution and signal quality, particularly for high-frequency activity, due to the distance of electrodes from the cortex [1]. In contrast, ECoG stands out with its unrivaled spatial and temporal resolution, achieved through small, closely spaced electrodes, enabling high-precision measurement of neural-population activity. ECoG's superior decoding capabilities are crucial for complex tasks like speech waveform reconstruction, as it can record activity from various brain areas simultaneously and with high quality [1]. This is especially relevant in scenarios such as multi-task classifications involving complex patterns like visual stimuli recognition, where ECoG has been observed to outperform EEG [14].

The efficiency of ECoG is also influenced by electrode deployment. The overall accuracy of ECoG systems improves with an increased number of electrodes, yet this improvement plateaus beyond eight regularly distributed electrodes. This highlights a balance in electrode distribution efficiency essential for optimizing ECoG's performance in various tasks [14].

Chronic ECoG safety and efficacy studies in animals demonstrate reliable long-term neural signal recording with minimal cortical tissue damage. These findings suggest the po-

tential for long-term stability and decoding capabilities of ECoG signals, crucial for chronic human BCI applications [1]. The major drawback of ECoG, however, is its invasiveness, requiring surgical procedures like burr holes or craniotomies for electrode placement, limiting its use to specific clinical scenarios or research studies [1].

In summary, ECoG emerges as a highly valuable tool in the realm of BCIs. Its exceptional spatial and temporal resolution, coupled with the ability to accurately decode a wide array of neural activities, positions it as an indispensable asset in advanced BCI research, notwithstanding its invasive nature and associated clinical limitations.

2.1.3 Non-Invasive Methods

2.1.3.1 Electroencephalography (EEG)

Electroencephalography (EEG) has established itself as a vital component in the realm of brain-computer interface (BCI) technology, showcasing significant advancements across diverse applications. In neuroprosthetics, EEG-based BCIs have revolutionized care for individuals with degenerative diseases like ALS, translating neuronal electrical signals into actions that enhance independence and mobility. This technology extends to communication aids for paralysis patients, where scalp-placed electrodes convert brain activity into synthesized speech, offering a new communication avenue. In the gaming industry, EEG's integration has led to innovative experiences; games such as MindRDR and Brain Wars utilize EEG sensors to measure players' emotional responses and concentration levels, thereby tailoring the gaming experience. The educational sector is not left behind, with EEG being used to provide real-time cognitive state feedback, aiding in personalized learning and stress reduction. Additionally, in mental healthcare, EEG's capability for real-time monitoring of brain activity presents novel treatment possibilities for conditions like depression, anxiety, and addiction [14].

EEG's application extends further into sleep medicine, where it plays a critical role in enhancing sleep quality and diagnosing sleep-related disorders [14]. The advent of advanced sensor technologies, including electrochemical transistor-based electrodes and nanotechnologies, has notably improved EEG's signal quality and practicality [15]. Neurofeedback-assisted BCIs, offering self-regulation and control over specific brain rhythms, have shown promising results in rehabilitation [15]. Despite these advancements, challenges persist, such as the necessity for personalized BCI designs tailored to individual brain functions and addressing physiological factors that impact BCI performance [18, 20]. Technological challenges, including developing more robust features for hybrid BCIs, cost and maintenance issues, and addressing socioeconomic concerns like ethical considerations and user safety, remain crucial areas for ongoing focus [20]. Key datasets such as DEAP, MAHNOB, DREAMER, GAMEEMO, and SEED, utilized by a majority of studies, along with advanced EEG systems like the Biosemi ActiveTwo, Emotiv Epoc headset, and Brain Vision LLC's Neuroscan Mobita, are fundamental in capturing high-fidelity EEG signals for emotion recognition research [18].

Recent advancements in BCI neurofeedback highlight its role in inducing brain plastic-

ity, aiding in the repair and control of nervous system outputs [20]. This aspect of EEG-based BCIs emphasizes their potential in neurorehabilitation, where they assist in the re-establishment of control through neural prostheses and reactivation of impaired synaptic networks [20]. Concurrently, significant strides in sensor design have emerged. Innovations like quasi-dry electrodes, which combine the advantages of both dry and wet electrodes, reflect a focus on improved signal-to-noise ratio, resolution, and user comfort [20]. Enhanced electrode technologies, such as transistor-based amplification and nanowire sensors, have markedly improved EEG signal quality [20]. Moreover, the advent of hybrid BCI systems, integrating modalities like SSVEP/ERP and SSVEP/MI, demonstrates a forward leap in BCI robustness and functionality [20].

2.1.3.2 Functional Near-Infrared Spectroscopy (fNIRS)

Functional near-infrared spectroscopy (fNIRS) stands out as a non-invasive brain imaging technique leveraging light to measure changes in oxygenated and deoxygenated hemoglobin levels [14]. This modality offers the capability to monitor both regional and global cortical activities, proving beneficial in a range of brain-computer interface (BCI) applications such as motor imagery, language processing, and affective state recognition [14].

The setup of fNIRS, involving the number and arrangement of sensors, is crucial for the accurate collection of data. These sensors allow for the monitoring of multiple brain areas simultaneously, providing excellent spatial resolution. However, fNIRS faces limitations in temporal resolution when compared to EEG, making it less suitable for tasks requiring rapid response detection [14]. Moreover, the signals obtained through fNIRS are relatively weak and prone to noise interference, posing challenges in data interpretation [14].

In terms of practical considerations, fNIRS systems are generally more costly than EEG setups and necessitate specialized training for effective use and interpretation of results. There are also safety considerations due to the exposure to light, which, if not properly managed, could lead to potential eye damage [14]. Despite these challenges, fNIRS remains a valuable tool in the realm of neuroimaging and brain research, offering unique insights into brain function and activity.

2.1.3.3 Magnetoencephalography (MEG)

Magnetoencephalography (MEG) is a non-invasive technique that measures magnetic fields generated by electrical currents in neurons, offering excellent temporal resolution [14] ($\approx 5,000$ Hz [21]). This feature makes MEG particularly useful for tracking dynamic neural activities related to cognitive processes such as attention and memory, as well as for various brain-computer interface (BCI) applications [14]. A key factor in MEG's accuracy is the density of sensors and their placement, which should be tailored to individual anatomy to optimize signal fidelity [14].

One of the major advantages of MEG is its good signal-to-noise ratio, providing strong and reliable signals that are beneficial for BCI applications [14]. It also offers high spatial resolution, enabling accurate mapping of brain activity [14]. However, the high cost of MEG

equipment limits its practicality outside specialized research laboratories and hospitals [14]. This cost factor, although contributing to its high accuracy, poses a significant barrier to more widespread use. Additionally, MEG is sensitive to head motion artifacts, which can interfere with the accuracy of the recorded signals [14]. It employs superconducting quantum interference devices (SQUID) for detecting magnetic fields, capitalizing on their high sensitivity to neural activity [15].

Despite its advantages, MEG-based BCI is still in its infancy and requires further research and exploration [15]. It offers superior spatiotemporal resolution compared to EEG, making it particularly useful in epilepsy treatment for identifying seizure foci and aiding in surgical planning [15]. However, its sensitivity to external environmental noise remains a challenge [15]. Overall, MEG represents a valuable tool in neuroimaging, providing unique insights into brain activity and function.

2.1.3.4 Functional Magnetic Resonance Imaging (fMRI)

Functional magnetic resonance imaging (fMRI) has emerged as a pivotal tool in neuroimaging, characterized by its superior spatial resolution, a feature instrumental in pinpointing active brain sections [15]. Operating at electromagnetic fields of 3 T or 7 T, fMRI is adept at measuring hemodynamic and metabolic activities within the brain, utilizing these parameters to identify areas of neural activation [15].

However, fMRI's application is constrained by several limitations. Notably, it suffers from poor temporal resolution, typically around 1-2 seconds (≈ 0.5 Hz), coupled with a physiological delay of 3-6 seconds due to the hemodynamic response [15, 21, 22]. This sluggish response hinders its suitability for real-time applications, a significant limitation in the context of brain-computer interface (BCI) technologies and rapid communication scenarios. Furthermore, fMRI's efficacy is compromised by its sensitivity to head movements, which can introduce artifacts into the imaging data [15].

Recent advancements in artificial intelligence have notably enhanced the decoding of brain activity from fMRI data, particularly in visual perception [21]. These innovations, while significant, have not overcome the intrinsic limitations of fMRI, such as its limited temporal resolution. Consequently, fMRI remains primarily focused on low-level visual features and faces challenges in fully exploiting its potential in real-time applications and dynamic stimulus decoding, such as speech and videos [21]. In summary, while fMRI stands out for its spatial resolution and cost-effectiveness compared to other modalities like PET and EEG, its temporal limitations and sensitivity to motion artifacts present significant challenges for its broader application in neuroscientific research and clinical settings [15, 21].

2.1.3.5 Positron Emission Tomography (PET)

Positron Emission Tomography (PET), a highly sensitive, non-invasive neuroimaging technique, is primarily employed to measure hemodynamic activity related to cerebral blood flow. This method uses radionuclides embedded in biological molecules, emitting positrons to enable the detection of diverse brain activities and abnormalities. PET's efficacy is partic-

ularly highlighted in its ability to localize specific brain regions, as demonstrated in studies involving hand grasping movements, showcasing its precision in neuroimaging applications. Additionally, its role extends to the molecular imaging of cancer and the detection of brain tumors [15].

In the realm of oncology, PET, especially through the use of ^{18}F -fluorodeoxyglucose (FDG), is indispensable for tumor segmentation. The quantification of metabolic activity via metabolic tumor volume (MTV) and total lesion glycolysis (TLG) in PET scans is critical for predicting patient outcomes and assessing treatment responses in various malignancies. These measures provide valuable insights into the metabolic activity of tumors, with higher metabolic activity in pretreatment scans often signaling a worse prognosis [23].

Beyond oncology, PET's clinical value also extends to diagnosing dementia in Alzheimer's patients, significantly contributing to the understanding of neurological disorders. This specific application of PET in Alzheimer's diagnosis underlines its broader clinical relevance beyond cancer imaging [15]. Notably, FDG PET's approach to measuring MTV offers distinct advantages over traditional anatomic imaging methods like MRI and CT. It is easier, faster, and potentially more reflective of the actual shrinkage of the viable tumor portion post-chemo- or radiotherapy [23].

However, the integration of MTV into standard clinical practice faces challenges, primarily due to the absence of a consensus on the optimal methods for tumor segmentation in FDG PET images. This lack of a definitive approach emphasizes the need for ongoing research and development in PET imaging for oncology [23]. In comparison to EEG, which primarily measures electrical brain activity, PET provides a contrasting perspective by focusing on metabolic processes. This difference is significant, as combining PET with EEG can yield a more comprehensive understanding of brain function and structure [17, 20].

Despite its advantages, PET's application is limited by certain drawbacks, notably its ineffectiveness in obtaining cross-sectional brain images and accurately pinpointing the exact locations of tumors or injuries. This limitation highlights a critical area where PET can be complemented by other imaging techniques [15, 20]. Additionally, factors like high operating costs and limited portability constrain PET's widespread use, particularly in contrast to more accessible techniques like EEG [20].

2.2 Classical Paradigms in BCI Systems

2.2.1 P300

The P300 paradigm is a prominent component in the realm of Brain-Computer Interface (BCI) systems, known for its high accuracy and ease of use. Developed in the 1980s, P300 leverages the EEG's ability to capture event-related potentials, particularly useful when a user recognizes specific stimuli. This process results in the emergence of the P300 wave, approximately 300 milliseconds post-stimulus, serving as an indicator of the user's cognitive engagement and attention. The practicality of P300-based BCIs extends to a wide array of applications, including medical rehabilitation, assistive communication, and cognitive as-

assessment, among others, thereby highlighting its versatility and significance in diverse fields [14].

However, the P300 paradigm is not without its challenges. One of the primary drawbacks is its susceptibility to noise interference, which can adversely affect the quality of data collected. Additionally, achieving accurate responses often necessitates a considerable number of trials, leading to slower response times. This characteristic, in turn, impacts the efficiency and user experience of P300-based BCIs [14].

Regarding the generation of the P300 response, a possible method involves Donchin's Speller. This system employs an alphanumeric matrix displayed on a computer screen. The stimuli are created by randomly flashing rows and columns of the matrix. When a user concentrates on a specific cell, the flashing of that cell's row or column becomes a significant event, eliciting the P300 response [4].

2.2.2 Steady-State Visual Evoked Potentials (SSVEP)

2.2.2.1 SSVEP Data Collection and Analysis

The Steady-State Visual Evoked Potentials (SSVEP) method, a cornerstone in EEG-based brain-computer interface (BCI) systems, relies on the rhythmic brain responses evoked by visual stimuli at specific flicker frequencies. SSVEP exploits these flicker frequencies from display monitors to generate EEG patterns that directly correlate with the user's gaze direction, thus enabling precise tracking of eye movements [14]. This capability is achieved as participants focus on changing visual stimuli, like colored squares, each eliciting a unique neural signature corresponding to the light's frequency. This method's robustness lies in its ability to accurately track eye movements without the need for additional equipment, such as cameras, making it a versatile tool in various applications ranging from virtual reality and AI research to medical diagnostics [14].

SSVEP-based BCIs are noted for their high information transfer rates and low training requirements, offering significant advantages over other BCI methods such as P300 in terms of response speed and bandwidth, crucial for real-time applications [2]. However, they also present challenges, including a lack of flexibility and a cumbersome setup process. These factors, especially the need for detailed calibration, can affect the portability and adaptability of SSVEP systems [14].

A notable example in SSVEP research is the Wang Open Dataset presented in 2017, which includes EEG data from 35 subjects, comprising both experienced and naive participants [16]. This dataset exemplifies the typical SSVEP experimental setup, where subjects are exposed to visual stimuli flickering at various frequencies. EEG data is initially recorded at a high sampling rate and later downsampled for analysis. Comprehensive datasets like this are vital for developing and validating SSVEP-based BCIs, offering a rich source of data for exploring SSVEP phenomena in diverse populations [16]. Other widely used datasets include the Nakanishi Open Dataset from 2015, and the BETA Open Dataset from 2020 [16].

2.2.2.2 Advancements in SSVEP Signal Processing

Recent advancements in SSVEP signal processing, particularly in the preprocessing phase, have significantly leveraged deep learning techniques [16]. Central to these advancements are the use of time-frequency transforms and filter banks, which play a pivotal role in enhancing the quality and interpretability of SSVEP signals. Filter bank analysis, involving sub-band decompositions with multiple filters of varying pass-bands, is especially crucial. The integration of techniques like the Filter Bank Canonical Correlation Analysis (FBCCA), which combines fundamental and harmonic frequency components, has markedly improved SSVEP detection over traditional methods [16]. Moreover, preprocessing steps like applying frequency filters to mitigate noise and implementing time-frequency transforms such as the Fast Fourier Transform (FFT) have proven effective in simplifying feature extraction. These methods have greatly aided deep learning models in efficiently processing SSVEP inputs, thereby making frequency features more accessible and enhancing overall model performance in SSVEP classification [16].

2.2.2.3 Deep Learning Models in SSVEP BCI

Deep learning models have become increasingly central in SSVEP-based BCIs. These models typically fall into three categories: fully connected neural networks, convolutional neural networks (CNNs), and recurrent neural networks (RNNs), including Long Short-Term Memory (LSTM) networks [16]. CNNs, known for their convolution layers, require less computational power and are less prone to overfitting, particularly with limited training data [16]. RNNs, and LSTMs specifically, are notable for their capacity to remember input sequences, making them suitable for applications where the temporal dynamics of SSVEP signals are critical [16].

2.2.3 Motor Imagery (MI)

Motor Imagery (MI), an important component in Brain-Computer Interface (BCI) research, involves users mentally simulating motor actions without physical movement. Rooted in the primary sensorimotor area of the brain, MI mimics the dynamics of actual movement [12]. Its diverse applications include controlling assistive devices like wheelchairs, enhancing rehabilitation, and aiding communication for those with speech impairments [14].

An integral aspect of MI is the understanding and leveraging of sensorimotor rhythms, specifically the mu (8-13 Hz) and beta (12-30 Hz) rhythms. When cerebral activity is associated with motor activity, variations in the amplitudes of these rhythms occur, which are essential for MI in BCIs since they are not related to actual movement. Notably, these sensorimotor rhythms are utilized in BCIs to interpret cognitive tasks such as prediction, learning, and multisensory integration [15].

MI research has been focusing on enhancing system accuracy and robustness through advanced techniques. Ensemble classifiers, particularly those using the sub-band common spatial pattern (SBCSP) method, are employed to address individual differences in rhythmic

patterns related to MI. Moreover, the incorporation of fuzzy integral and particle swarm optimization (PSO) techniques is used for regulating individual differences in parameters, thus enhancing the system's overall performance [12]. However, challenges such as distinguishing between real and imagined motor tasks and user fatigue impact the performance and accuracy of MI systems [14]. To enhance MI classification accuracy and speed, extensive research has been focused on applying various machine learning algorithms such as KNN, LDA, SVM, and ANN [17].

In addition to MI, Slow Cortical Potentials (SCPs) and non-motor cognitive tasks play significant roles in BCI. SCPs, characterized by slow voltage shifts in EEG signals, lasting several seconds, are indicative of cortical activity variations and are employed in various BCI control applications, although they require extensive training and usually yield lower information transfer rates (ITRs) [15]. Non-motor cognitive tasks, such as visual counting and mental rotation, improve the performance of adaptive BCI systems for users with severe motor impairments [15]. In MI-based BCIs, channel and feature selection are crucial, with techniques like modified Stockwell transforms and multilevel hybrid PSO-Bayesian linear discriminant analysis being employed for signal classification and optimization. These methods reduce the number of features needed and shorten testing time, thus enhancing the practicality of MI-BCI systems, particularly for patients with motor disabilities [11].

2.3 BCI Signal Processing Techniques

2.3.1 Path Signature

Path Signature is an innovative and robust feature for Brain-Computer Interface (BCI) signal processing, offering a novel approach to interpreting electroencephalography (EEG) signals. As a sequence of iterated integrals derived from a multidimensional path, the path signature uniquely maintains invariance to translation and time reparametrization. This robustness is especially advantageous for EEG time series, which frequently suffer from non-stationarity and environmental noise [25].

The application of path signature in BCI is driven by its ability to capture geometric features of the path. The first and second levels of the signature offer an intuitive understanding by focusing on the displacement of the path and the relationships between pairs of path components, respectively. These initial levels are effective in capturing the cyclic order of EEG data, which proved to be highly valuable in both intra- and inter-subject classifications. The higher levels of the signature add further analytical depth by detailing more complex features of the EEG signals, thus transforming nonlinear classification challenges into more manageable linear ones [25].

In BCI systems, the path signature method stands out for its ability to handle the high variability inherent in EEG signals. This variability, which stems from differences in sessions and users, poses a significant challenge in establishing robust classifiers. The path signature method addresses this issue by providing features that are more resistant to these variances. By combining the path signature with standard classifiers like the Riemannian classifier,

which leverages the geometric structure of symmetric positive definite matrices, there is a noticeable improvement in performance, particularly in scenarios involving noisy and low-quality data [25].

The real-world impact of this method was evident in studies that employed the Path Signature as a feature map for EEG signal classification. Utilizing n -dimensional paths derived from EEG data, the Path Signature method, particularly at its second level, demonstrated effective results. This was evident in datasets like the BCI competition IV 2a, where it achieved high accuracy in both intra- and inter-subject classifications. Particularly notable is its success with subjects less responsive to traditional classifiers, indicating its potential in addressing inter-subject variability and reliable EEG signal interpreting [25].

2.3.2 Dimensionality Reduction

Dimensionality reduction in Brain-Computer Interface (BCI) systems, particularly through Independent Component Analysis (ICA) and Principal Component Analysis (PCA), is important for enhancing EEG signal processing. ICA is instrumental in segregating mixed signals into independent sources, thereby facilitating the extraction of critical information like event-related potentials. Its utility in artifact removal is significant, reducing false positives caused by external interferences such as movement artifacts [14]. However, ICA algorithms face challenges, including ambiguity in source origin and reduced efficacy with small datasets.

PCA focuses on transforming correlated observations into a set of uncorrelated variables. This method excels in noise reduction and data dimensionality reduction, though it does not inherently ensure effective classification. By eliminating noise through the identification of small eigenvalues, PCA partially denoises the EEG data, contributing to cleaner signal interpretation [15, 17]. The integration of PCA with other methods like Ensemble Empirical Mode Decomposition (EEMD) enhances EEG data quality, particularly in artifact suppression [17].

In addition to ICA and PCA, other advanced techniques like EEMD and Multivariate Adaptive Moving Average–Empirical Mode Decomposition (MAMA-EMD) have emerged. EEMD, when combined with techniques like Canonical Correlation Analysis (CCA), enhances the number of channels available for ICA by leveraging interchannel information and addressing challenges in EEG data with low signal-to-noise ratio [17]. MAMA-EMD, an advancement over EEMD, focuses on extracting spikes into the first Intrinsic Mode Function (IMF) to improve the accuracy of ensuing IMF screening and mitigate mode-splitting effects, although it may encounter difficulties with multi-point spikes [17]. Additionally, CCA is proficient at extracting inter-band correlation (IBC) features based on differential entropy (DE), providing insights into the mutual interactions between EEG frequency bands under various emotional expressions. This method is particularly beneficial as it avoids the need for surrogate testing, thus reducing computational complexity. In practice, the combination of IBC features with traditional DE features at the decision level has been shown to significantly enhance the accuracy of emotion recognition [24].

These dimension reduction techniques, including ICA, PCA, EEMD, and MAMA-EMD, are essential in refining BCI systems. They offer distinct yet complementary approaches, contributing to more accurate and effective EEG signal processing in BCI research. While

ICA and PCA are adept at identifying independent sources and reducing data noise, respectively, EEMD and MAMA-EMD bring additional capabilities in dealing with complex artifacts and improving signal clarity.

2.3.3 Wavelet Transformations (WT)

Wavelet Transformations (WT) play a crucial role in Brain-Computer Interface (BCI) systems. WT uniquely decomposes EEG data into frequency components, which is pivotal for analyzing brain signals associated with various cognitive tasks [14]. This decomposition is essential in understanding how different frequencies in the brain relate to specific mental activities, enhancing the capability of BCIs to interpret and respond to user inputs effectively.

Specifically, the Discrete Wavelet Transform (DWT) is employed to calculate the scaling and detail components of a waveform, crucial for EEG signal processing. DWT's ability to handle artifact removal through the thresholding of decomposed coefficients makes it a valuable tool in EEG analysis. However, it faces challenges when the spectral properties of the signal overlap with those of the artifacts [14]. To overcome the limitations of DWT, the Stationary Wavelet Transform (SWT) is used. SWT eliminates temporal dispersion, a drawback of DWT, and allows for the tracking of changes in the harmonic components of EEG signals over time [14].

In the broader context of feature extraction in BCIs, Wavelet Transformations stand alongside other methods such as Autoregressive Modeling, Continuous and Discrete Wavelet Transforms, and Matched Filtering. Each of these techniques offers unique advantages for processing and analyzing EEG signals, contributing to the development of more accurate and responsive BCI systems [15].

2.3.4 Hidden Markov Models (HMM)

Hidden Markov Models (HMMs) are integral to BCI systems, primarily for recognizing user intentions through EEG signal analysis. HMMs, based on Bayesian posterior probability maximization, are adept at classifying time series data, making them suitable for tasks like movement intent detection or mental arithmetic classification. Central to their functionality are three core problems: evaluation (determining the probability of a sequence of visible states generated by the model), decoding (identifying the most probable sequence of hidden states), and learning (optimizing model parameters from training data). Techniques like the Viterbi algorithm and Baum-Welch algorithm are employed for these purposes. Despite their proven effectiveness in classifying EEG time series and the capability to calculate likelihoods of feature vector sequences, HMMs face challenges in BCI application. A significant hurdle is the identification of stable observable states within EEG signals, a task complicated by the intricate nature of brain activities and signal dynamics. This limitation hinders their widespread adoption in BCI development, suggesting a need for further exploration into EEG signals and neurophysiological brain functioning to enhance their applicability [14].

2.3.5 Artificial Neural Networks (ANN)

Artificial Neural Networks (ANN) are particularly adept at processing and classifying brain signals in EEG-based systems. ANNs exhibit excellent flexibility and are available in numerous architectures, including Probabilistic Neural Networks (PNN), Fuzzy ARTMAP ANN, Finite Impulse Response Neural Networks (FIRNN), and Probability Estimating Guarded Neural Classifiers (PeGNC). These diverse architectures enable ANNs to enhance their adaptability to various signal processing tasks [15].

Convolutional Neural Networks (CNNs) are notable for their efficiency in managing complex tasks like image classification within EEG data. By employing spectrograms as inputs, CNNs simplify the classification process. Furthermore, specific architectures like ShallowNet, which employ time convolution and spatial filtering, mirror the capabilities of traditional methods such as the Filter Bank Common Spatial Pattern (FBCSP). This parallelism allows for enhanced processing of EEG signals, thereby facilitating accurate signal classification and supporting the rapid decoding of complex signals such as hand gestures in electrocorticography [14].

However, the successful implementation of these neural networks in EEG-based systems is not without challenges. One significant hurdle is their dependence on the quantity and quality of available data. The performance of deep learning methods, like neural networks, improves with access to more data, leading to better generalization and accuracy in signal interpretation [14]. This data dependency necessitates the use of data augmentation techniques to address the issue of limited data availability. Augmentation methods artificially expand the training dataset, increasing variability and the number of samples. This expansion is crucial for training robust neural network models that can effectively handle the complexities of EEG signal processing [14].

2.3.6 Genetic Algorithms (GA)

In the realm of Brain-Computer Interface (BCI) systems, the application of Genetic Algorithms (GA) has significantly advanced feature selection and classification processes. Notably, GA has been effectively used in Steady State Visually Evoked Potential (SSVEP)-based BCIs for optimizing feature selection, enhancing classification accuracy, and reducing the dimensionality of data [2]. Similarly, in motor imagery-based BCI systems, GA coupled with Stockwell transform and Bayesian linear discriminant analysis has shown to significantly improve classification performance by selecting the most relevant feature sets, achieving high sensitivity, specificity, precision, and accuracy [3]. Furthermore, a study on P300-based BCIs demonstrated the efficacy of GA in significantly enhancing classification performance and reducing feature count, suggesting its potential in portable BCI configurations [4]. Additionally, a notable recent study employed Multiclass Genetic Programming with Multidimensional Populations (M3GP), significantly enhancing emotion recognition from EEG data [8].

2.3.7 Particle Swarm Optimization (PSO)

Particle Swarm Optimization (PSO) has emerged as a pivotal technique in advancing Brain-Computer Interface (BCI) technology, particularly in the realm of Motor Imagery (MI) and emotion recognition. In the context of MI-BCI systems, the study referenced at [11] introduces a novel Multilevel PSO (MLPSO) framework, which significantly optimizes channel and feature selection. This approach, characterized by its simple computation and rapid convergence, demonstrates exceptional improvements in classification accuracy, achieving up to 99% accuracy with a substantially reduced feature set. This reduction in features directly translates into enhanced efficiency and potential for real-time application, a critical aspect for practical BCI systems.

Complementing this, the research at [12] explores the integration of fuzzy integral with PSO to address variability in EEG-based BCIs. This approach results in a more robust and stable system, adept at processing uncertain and imprecise EEG signals, particularly during MI tasks. The fusion of fuzzy integral with PSO optimizes classifier confidence levels, enhancing overall system performance. This is exemplified in the study's successful demonstration of effective real-time control of a robotic arm. Additionally, the study at [13] employs a modified PSO method with a multi-stage linearly-decreasing inertia weight (MLDW-PSO) for emotion recognition from EEG signals. This adaptation allows for more nuanced control and improved accuracy in feature selection, highlighting PSO's versatility and efficacy in different BCI applications. These studies collectively underscore the transformative potential of PSO in BCI, especially in optimizing feature selection and classifier performance for both motor imagery and emotion recognition applications.

2.4 EEG Recognition Tasks

2.4.1 EEG Text Recognition

In the paper at [10], the researchers employed the ZuCo 1.0 and 2.0 datasets for their experiments. These datasets recorded text and EEG corpus during Normal Reading (NR) and Task-Specific Reading (TSR) tasks, using a 128-channel system at a 500 Hz sampling rate. After noise canceling, 105 channels were used for translation. For word-level EEG feature translation, the DeWave model utilized eye-fixation data associated with each word during reading, available in the ZuCo dataset. The EEG waves corresponding to words fixated upon multiple times were concatenated for processing. The statistical results of the EEG wave segments were computed across four frequency band filters: Theta (5-7Hz), Alpha (8-13Hz), Beta (12-30Hz), and Gamma (30Hz and above), resulting in a feature size of 840 for each word. For decoding purposes, the raw EEG signals were normalized to a range between 0 and 1.

DeWave's training involved a multi-stage process. Initially, the language model wasn't included in weight updates, focusing on training a proper encoder projection and a discrete codex representation for the language model. In the subsequent stage, the weights, including those of the language model, were fine-tuned. The model implementation for word-level

EEG features involved using 56 tokens, each with an 840 embedding size. For raw EEG waves, the signals were clipped or padded up to 5500 sample points. A transformer layer with eight heads and a 1×1 convolutional layer fused multiple EEG channels into an embedding sequence of size 512. DeWave used a codex of size 2048, where each latent codex was an embedding of size 512. The training process involved different learning rates and epochs for self-supervised decoding for raw waves and for training the codex. All models were trained on Nvidia V100 and A100 GPUs. Performance evaluation was based on Natural Language Processing (NLP) metrics like BLEU and ROUGE. DeWave's results were compared with existing EEG-to-Text methods, and its performance was benchmarked against baselines established for raw EEG waves. This comparison involved segmenting entire EEG waves into a sequence embedding and using Wave2Vec, adapted for brain waves. Additionally, unsupervised raw EEG waves classification methods like BENDR and SCL were used for comparison to underscore the impact of discrete encoding.

In summary, the DeWave framework introduces a novel approach to EEG-to-text translation, demonstrating its effectiveness through robust performance metrics. The use of discrete encoding and a multi-stage training process highlights the model's innovative approach to translating brain dynamics into text.

2.4.2 EEG Emotion Recognition

In this section, we explore methodologies from the realm of EEG-based emotion recognition, encompassing various studies utilizing techniques such as deep learning, genetic programming, and advanced signal processing for emotion classification. Study [6] presents a model integrating deep convolutional networks with type-2 fuzzy activation functions and GANs, achieving over 98% accuracy in emotion classification. The research in [7] employs fused Connectivity Feature Maps and a CNN, finding the highest accuracy in Gamma sub-band classifications. In [8], the application of Multiclass Genetic Programming with Multidimensional Populations enhances feature selection, leading to a significant improvement in classification rates. Study [9] introduces a hybrid deep learning model, combining CNN and LSTM RNN, to efficiently process both spatial and temporal EEG data, while study [13] leverages Particle Swarm Optimization for effective feature selection, demonstrating its efficacy in both offline and online emotion recognition scenarios. Study [26] proposes the Joint Contrastive Learning framework with Feature Alignment (JCFA), which improves cross-corpus EEG-based emotion recognition by aligning time- and frequency-based embeddings and incorporating spatial features using a graph convolutional network. Additionally, the SleepEGAN model [34] introduces a GAN-enhanced ensemble deep learning model for addressing class imbalance and individual heterogeneity in EEG signal processing. These studies collectively contribute to advancing EEG-based emotion recognition, offering insights into the potential of various computational approaches in interpreting complex emotional states through EEG signals.

2.4.2.1 Data Acquisition and Preprocessing

In the realm of EEG-based emotion recognition, meticulous data acquisition and preprocessing techniques are paramount for obtaining reliable and accurate results. This is evident in the study by [6], where EEG signals were captured using a 21-channel EEG device, adhering to the 10–20 system standard. Notably, the dataset comprised data from eleven volunteers, encompassing a diverse age range (18–32 years) and gender distribution. The utilization of specific electrodes (A1 and A2) for referencing brain signals underscores the precision in data collection. Furthermore, the selection of 12 active electrodes, including Pz, T3, C3, among others, was critical in ensuring robust signal acquisition, particularly for processing emotional stimuli induced by musical elements. Similarly, in the context of class imbalance and individual heterogeneity, [34] introduces a GAN-enhanced ensemble deep learning model for EEG signal processing, addressing similar challenges in sleep stage classification.

The preprocessing phase in [6] involved sophisticated techniques like the application of a notch filter for eliminating power supply frequency interferences and a Butterworth filter within the 0.05 to 45 Hz range, optimized for emotion induction from EEG signals. Similarly, in the research documented in [7], the DEAP dataset was employed, which is renowned for its comprehensive EEG and peripheral physiological signals from 32 subjects. The use of the BioSemi ActiveTwo system, in compliance with the 10/20 international standard, further reinforces the rigor in data acquisition methodologies. This study's focus on the Gamma and Beta sub-bands of EEG signals for emotion recognition provides a nuanced understanding of the EEG's frequency components in relation to emotional states.

Moreover, the paper [9] delves into the broader context of emotion-related feature extraction from EEG signals, highlighting the significance of frequency domain analysis, especially utilizing Fast Fourier Transform (FFT). This method is integral for decomposing EEG signals into frequency bands, each potentially correlating with different emotional states. In contrast, the study in [8] introduces an approach employing the Biometric EEG dataset (BED), collected using the Emotiv EPOC+ headband. This dataset's tailoring for emotional state analysis exhibits the evolving landscape of EEG data acquisition tools.

The study [13] further adds to the discourse by leveraging the DEAP dataset for EEG emotion recognition. This study stands out for its focus on a diverse array of EEG features, including Hjorth features, fractal dimension features, and wavelet features. The multi-stage linearly-decreasing weight (MLDW-PSO) method introduced for feature selection is a testament to the continuous evolution of preprocessing strategies in EEG-based studies.

2.4.2.2 Relevant EEG Electrodes

The exploration of EEG electrodes in emotion recognition studies is crucial in understanding the neural underpinnings of emotional processes, with a focus on both spatial resolution and frequency-specific brain activity. Studies have demonstrated the importance of specific electrode placements and frequency bands in the context of emotional state analysis, underscoring the complexity of EEG data in emotion recognition.

In the study referenced as [6], researchers utilized a 21-channel EEG device, adhering to the 10-20 standard, for recording brain signals. Importantly, 12 active electrodes, including Pz, T3, C3, C4, T4, F7, F3, Fz, F4, F8, Fp1, and Fp2, were identified as key in the processing of EEG signals for emotion recognition. This selection of electrodes underscores the significance of spatial resolution in EEG studies and points towards specific brain regions relevant for emotional processing. Similarly, the study denoted as [7] employed the BioSemi ActiveTwo system, in line with the 10/20 international standard, to record EEG signals. This standardization in electrode placement is pivotal for ensuring comparability across studies and highlights the relevance of specific brain regions in the context of emotion recognition. The study's focus on the Gamma and Beta sub-bands of EEG signals further emphasizes the importance of considering frequency-specific brain activity in relation to different emotional states.

In contrast, the research in [8] utilized the Emotiv EPOC+ headband for EEG data collection in their study of emotion recognition. The use of this cost-effective EEG recording tool exemplifies the diversification in EEG data collection methods. While it may offer less spatial resolution compared to more extensive electrode setups, it highlights the growing accessibility and applicability of EEG technology in various contexts, including emotion recognition.

The study at [9] delves into the broader context of emotion-related EEG signal analysis, emphasizing the importance of frequency domain analysis, particularly the Fast Fourier Transform (FFT), in decomposing the EEG signal into multiple frequency bands. This process facilitates the association of different emotional states with specific frequency bands, such as pleasant emotions with the frontal midline theta frequency band and experiences of anger with the beta-2, beta-3, and gamma frequency bands in the frontopolar electrodes. This nuanced understanding of the relationship between EEG signals and emotional states is crucial in developing more accurate and sophisticated emotion recognition systems. Lastly, the study indicated as [13] leverages Particle Swarm Optimization (PSO) for feature selection from EEG data, specifically focusing on high-dimensional features, including those derived from various electrode placements.

2.4.2.3 EEG Signal Preprocessing

In the domain of EEG-based emotion recognition, the process of signal preprocessing emerges as a critical component, serving as the cornerstone for enhancing data quality and interpretability. This significance is exemplified across various studies, each adopting meticulous methodologies to address the unique challenges posed by EEG data.

Central to this effort is the attenuation of extraneous frequencies and interference, as demonstrated by studies [6], [7], and [26]. In study [6], a 21-channel EEG device aligned with the 10–20 standard was employed, incorporating a notch filter to eliminate 50 Hz frequency interference from the power supply. Complementing this approach, study [7] utilized the BioSemi ActiveTwo system, also adhering to the 10/20 standard, while specifically focusing on Gamma and Beta EEG sub-bands. Additionally, study [26] introduced a contrastive learning strategy in the preprocessing phase to align time and frequency embeddings, important for refining data for subsequent analysis in cross-corpus scenarios. This targeted ap-

proach in preprocessing underscores the importance of isolating frequencies most relevant to emotion analysis, a practice crucial in refining data for subsequent analysis. Additionally, [34] employed a GAN architecture tailored for EEG signal generation, effectively balancing training samples among different classes to enhance classification performance.

Further emphasizing the role of preprocessing in tailoring EEG datasets for specific applications, studies [8] and [9] delve into spectral feature analysis. Study [8], utilizing the Emotiv EPOC+ headband, highlights the use of mel-frequency cepstral coefficients and autoregression reflection coefficients in the Biometric EEG dataset, specifically curated for emotion analysis. Complementarily, study [9] explores the intricate relationships between various EEG frequency bands and emotional states, underscoring the necessity of sophisticated preprocessing methods in delineating these complex interactions.

The innovation in preprocessing methods is also evident in study [13], which introduces Particle Swarm Optimization for feature selection in the realm of emotion recognition. This approach effectively addresses the high-dimensional nature of EEG data. By selecting EEG features pertinent to emotion recognition, such as Hjorth features and power spectral density, the study adeptly reduces data redundancy and enhances the accuracy of emotion classification.

By ensuring that the data entering the feature extraction phase are of the highest quality, these methodologies lay a solid foundation for more effective and precise emotion recognition, thus setting a benchmark for future research in the field.

2.4.2.4 Feature Extraction and Enhancement Techniques

The efficiency of feature extraction and enhancement techniques plays a critical role in advancing classification accuracy. Notably, the research in [6, 9] exemplify the integration of innovative methods to tackle the complexities of EEG signal processing. Study [6] utilizes Generative Adversarial Networks (GANs) for data augmentation, addressing overfitting issues, and employs preprocessing methods such as a notch filter and a second-order Butterworth filter. This approach underscores the significance of effective emotional induction through EEG. Complementing this, study [9] offers a comprehensive analysis of EEG feature extraction, encompassing time-domain, frequency-domain, and time-frequency domain features. It demonstrates the use of Fast Fourier Transform (FFT) and Short-Time Fourier Transform (STFT), as well as discrete wavelet transform (DWT), underscoring the multifaceted nature of EEG signal processing. The introduction of a hybrid model combining a Convolutional Neural Network (CNN) with a Long Short-Term Memory (LSTM) Recurrent Neural Network in this study marks a significant step in integrating spatial and temporal information for emotion recognition. Similarly, [34] utilized a combination of convolutional layers and LSTM for feature extraction in EEG data, enhancing the generalization and parsimony of the model.

Furthermore, the studies [7] and [8] explore the potential of enhancing information content in EEG signals without increasing data size. The research in [7] effectively uses the Database for Emotion Analysis using Physiological Signals (DEAP) and introduces Connectivity Feature Maps (CFMs) by fusing various connectivity methods like the Pearson Cor-

relation Coefficient (PCC) and Phase Locking Value (PLV). This innovation focuses on the Gamma and Beta sub-bands of EEG signals, indicating a nuanced understanding of the frequency domain in relation to emotional states. In contrast, study [8] employs the Biometric EEG dataset (BED) and introduces a novel method involving Multiclass Genetic Programming with Multidimensional Populations (M3GP) for feature enhancement. This method, using spectral features such as mel-frequency cepstral coefficients (MFCC) and autoregression reflection coefficients (ARRC), exemplifies an advanced approach in feature selection, enhancing classification accuracy.

Lastly, the study denoted as [13] highlights the application of Particle Swarm Optimization (PSO), particularly a modified variant known as multi-stage linearly-decreasing weight (MLDW-PSO). This technique demonstrates an effective balance between global exploration and local exploitation in feature selection. Its successful application in both offline and on-line experiments with the DEAP dataset underscores the potential of advanced optimization techniques in refining feature selection for EEG-based emotion recognition systems.

The continuous development and optimization of these feature extraction and enhancement techniques are instrumental in creating more sophisticated, accurate, and efficient emotion recognition models.

2.4.2.5 Model Development and Optimization

A common thread among various studies is their focus on improving computational techniques and neural network architectures, aimed at enhancing the accuracy and applicability of emotion recognition systems.

One such approach is the employment of Generative Adversarial Networks (GANs) for data augmentation to mitigate overfitting, as presented in [6]. This study utilized a 21-channel EEG device, focusing on 12 active electrodes and employing preprocessing techniques like a notch filter and a second-order Butterworth filter. Its convolutional model architecture, featuring type-2 fuzzy activation functions and a Fully Connected (FC) layer with nonlinear Softmax activation, optimized via Stochastic Gradient Descent (SGD), achieved an impressive accuracy of over 98% in classifying emotions. The SleepEGAN model [34] integrates GANs for data augmentation and employs an ensemble learning strategy to address class imbalance and individual heterogeneity in EEG signals, achieving superior classification performance.

Similarly, the study in [7] uses the Database for Emotion Analysis using Physiological Signals (DEAP) and emphasizes the significance of EEG signal sub-bands, particularly Gamma and Beta bands, in emotion recognition. It introduced a fusion of different connectivity methods, like Pearson Correlation Coefficient (PCC) with Phase Locking Value (PLV), to enhance Connectivity Feature Maps (CFMs) without increasing their size. The study's Convolutional Neural Network (CNN) architecture, including ReLU activation function and batch normalization, optimized with the Adam algorithm, represents another example of methodological robustness in this field.

Additionally, the integration of deep learning with advanced machine learning techniques is a growing trend, as seen in [8]. This study, utilizing the Biometric EEG dataset (BED)

and Emotiv EPOC+ headband, adopted a novel approach by combining deep learning with Multiclass Genetic Programming with Multidimensional Populations (M3GP). This dual strategy in feature extraction, involving spectral features and autoregression reflection coefficients (ARRC), and the use of a one-dimensional convolutional neural network structure, achieved a mean classification rate of 92.1%, showcasing the potential of such integrative approaches.

The study in [26] introduces the JCFA framework, which includes a joint contrastive learning strategy to align time- and frequency-based embeddings without the need for labeled data, enhancing the model's generalizability across different datasets. The model's fine-tuning stage leverages spatial features using a graph convolutional network, further improving the accuracy of emotion recognition in cross-corpus scenarios.

Lastly, addressing challenges in high-dimensional data and real-time application, the study in [13] presented a novel approach using Particle Swarm Optimization (PSO) for EEG-based emotion recognition. This study introduced a modified PSO method, termed multi-stage linearly-decreasing weight (MLDW-PSO), adept in function optimization and feature selection. Through the use of EEG features extracted from various domains, this method achieved significant accuracy in both offline and online experiments on the DEAP dataset, further underscoring the potential of optimized feature selection strategies in this realm.

2.4.2.6 Performance Analysis and Application

One approach employing a 21-channel EEG device, integrating Generative Adversarial Networks (GANs) for data augmentation and deep convolutional networks with type-2 fuzzy activation functions, achieved over 98% accuracy in classifying positive and negative emotions [6]. This highlights the potential of sophisticated techniques in brain-computer interface applications. In another approach, SleepEGAN [34] demonstrated significant improvements in classification accuracy by generating high-quality samples for minority classes and utilizing ensemble learning to enhance robustness against individual heterogeneity.

In contrast, another study focused on Convolutional Neural Networks (CNNs) and Connectivity Feature Maps (CFMs), emphasizing the performance of different EEG sub-bands. It was found that the Gamma sub-band yielded higher accuracy in emotion classification compared to Alpha and Beta sub-bands. Mutual Information (MI) outperformed methods like Transfer Entropy (TE) in test set accuracy, despite TE's faster loss convergence. The study also explored subject-dependent and subject-independent performance, achieving 91.29% accuracy for Valence and 91.66% for Arousal classification using the PLV+MI method in the fused CFM approach, underscoring the challenge of consistent accuracy across different subjects [7].

Another novel method involved Multiclass Genetic Programming with Multidimensional Populations (M3GP) using the Biometric EEG dataset (BED), focusing on spectral features and a one-dimensional convolutional neural network, leading to a mean classification rate of 92.1% [8]. This demonstrated how advanced machine learning techniques can refine EEG-based emotion recognition processes.

Further contributions to this field include the integration of frequency domain and spa-

tial features of emotional EEG signals into a hybrid model combining a CNN and a Long Short-Term Memory (LSTM) Recurrent Neural Network (RNN), which achieved an average classification accuracy of 92.98% across 32 subjects [9]. This emphasized the importance of incorporating both spatial and temporal information in EEG data analysis.

Of note is the study [26], which demonstrates superior performance compared to existing SoTA methods on cross-corpus datasets. JCFA achieved a recognition accuracy of 67.53% for SEED-IV $\rightarrow$ SEED and 62.40% for SEED $\rightarrow$ SEED-IV, surpassing existing methods by an average accuracy increase of 4.09%. This highlights the potential of joint contrastive learning in enhancing emotion recognition accuracy, particularly for challenging negative emotions.

Lastly, a novel approach using Particle Swarm Optimization (PSO) for feature selection was introduced. The modified PSO method, named multi-stage linearly-decreasing weight (MLDW-PSO), proved effective in both offline and online experiments, achieving an average accuracy of 76.67% for four-class emotion recognition and 89.5% in an online two-class emotion recognition system [13].

Chapter 3

Methodology

3.1 Baseline Datasets

The SEED and SEED-V datasets, detailed in [27, 28] and [29], are comprehensive EEG-based emotion recognition datasets utilized in this research. Both datasets consist of EEG data collected using a 62-channel ESI NeuroScan System and were downsampled to 200 Hz. Preprocessing for both datasets involved bandpass filtering (0-75 Hz for SEED, 1-75 Hz for SEED-V) to remove noise and artifacts. Eye-tracking data were collected using SMI eye-tracking glasses.

3.1.1 SEED

The SEED dataset, as described in [27, 28], includes data from 15 Chinese subjects who participated in experiments designed to elicit positive, neutral, or negative emotions through 15 carefully selected film clips. The EEG data was collected and downsampled to 200 Hz. Preprocessing involved bandpass filtering (0-75 Hz) and segmenting the data to match the duration of each film clip, resulting in 45 files per subject. Each subject underwent three experimental sessions, spaced approximately one week apart, yielding data arrays corresponding to each of the 15 trials per session. The dataset also provides extracted features such as differential entropy (DE), differential asymmetry (DASM), and rational asymmetry (RASM), along with smoothed versions of these features. The SEED_Multimodal part includes raw EEG data, DE features, and eye-tracking data.

For our purposes, only the preprocessed EEG data from the SEED_EEG part was utilized. This approach excludes the generated features (DE, DASM, RASM) and the multimodal components, focusing solely on the preprocessed, segmented EEG signals for emotion recognition research.

3.1.2 SEED-V

The SEED-V dataset, detailed in [29], includes data from 20 subjects (10 males and 10 females) from Shanghai Jiao Tong University. Each subject participated in the experiment three times, with at least three days between sessions, watching 15 film clips per session

designed to elicit specific emotions: happiness, neutrality, sadness, disgust, and fear. The EEG data were downsampled to a 200 Hz sampling rate and processed with a bandpass filter between 1 Hz and 75 Hz. Differential entropy (DE) features were extracted across five frequency bands (delta, theta, alpha, beta, gamma). Eye movement features were also collected, including parameters such as pupil diameter, fixation, saccade, and blink.

For our purposes, only the preprocessed EEG data from SEED-V was utilized, specifically the raw EEG signals that had been downsampled and passed through a bandpass filter. This approach excludes the use of extracted features such as differential entropy (DE) and the eye movement data, ensuring a streamlined focus on the bandpass-filtered EEG signals for emotion recognition research.

3.2 EmotionWave Custom Dataset Creation

3.2.1 Participant and Experimental Setup

This study involved the creation of a custom dataset, hereafter referred to as EmotionWave, which involved a single participant, the researcher, who collected the data. The participant underwent three separate sessions, each conducted on different days with at least three days between sessions. Each session consisted of 15 trials, with each trial designed to elicit one of three emotions: positive, negative, or neutral. The EEG data were processed with a bandpass filter ranging from 0.5 Hz to 75 Hz and downsampled from the original 256 Hz to 200 Hz using polyphase filtering.

Overall, after segmenting the EEG data into 1-second windows with dimensions 14×200 , the dataset comprises 5760 EEG segments. Each segment represents a 1-second snapshot of the brain's activity during the emotional elicitation trials, providing a rich dataset for further analysis and model training.

3.2.2 Stimuli and Procedure

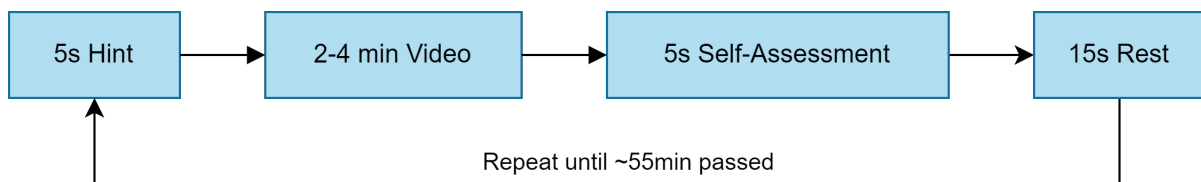


Figure 3.1: Diagram of the trial structure within a session.

The stimuli consisted of carefully selected video clips designed to induce specific emotions. Each trial followed a structured procedure, as illustrated in Figure 3.1, and described below:

- **5s Hint:** A brief hint indicating the emotion intended to be induced.
- **2-4 min Video:** The participant watched a film clip intended to elicit a specific emotion, with the EEG headset on.

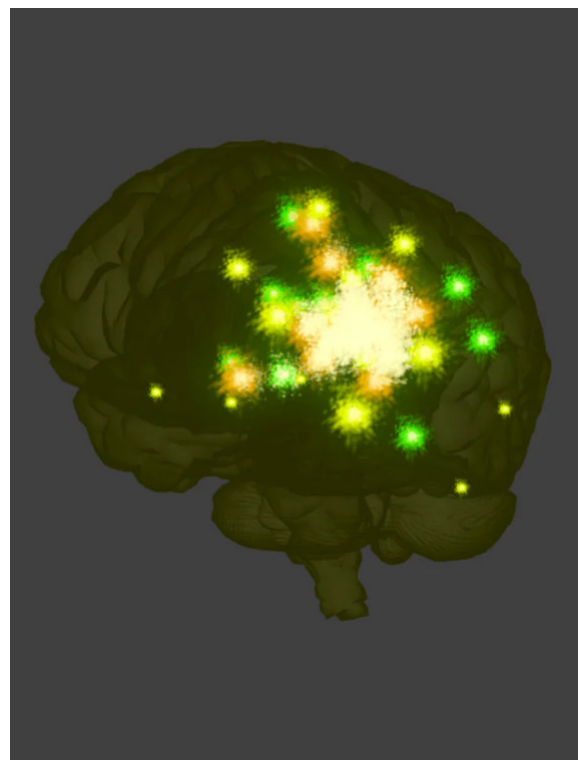
- **5s Self-Assessment:** Following the viewing of each film clip, the participant conducted a self-assessment to rate their emotional reaction. They used a scale from 1 to 10, where a rating near 10 indicated that the emotion was strongly conveyed and felt (e.g., feeling happy after a positive video), while a rating near 1 indicated the opposite. For neutral stimuli, a score of 10 was given if the participant's mood remained unchanged, and a score of 1 if their mood fluctuated.
- **15s Rest:** The participant was given a 15-second rest period in which they were instructed to close their eyes until the next hint appeared. This ensured a clear transition and preparation for the subsequent stimulus.

The trials were combined into sessions of about 50-55 minutes each to ensure the participant did not become tired. The order of the film clips was arranged to avoid presenting two clips eliciting the same emotion consecutively, thereby maintaining the participant's engagement and ensuring the effectiveness of emotion induction.

3.2.3 EEG Data Collection



(a) EMOTIV EPOC X headset.



(b) Brain activity visualization using BrainViz software. Colored points indicate activation regions, with varying intensities representing EEG signal strengths.

EEG data were collected using the EMOTIV EPOC X headset (Figure 3.2a). Brain activity patterns were visualized using BrainViz software (Figure 3.2b). The specifics of the EEG data collection process are as follows:

- **EEG Channels:** The headset has 14 channels positioned at AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4. These channels correspond to standard 10-20 system placements, ensuring comprehensive coverage of the scalp for effective data collection.
- **Reference Points:** Two reference points, CMS/DRL (located at P3/P4), and alternative reference points at the left/right mastoid process are used for grounding and noise reduction.
- **Sampling Rate:** The original sampling rate was set to 256 Hz, providing high-resolution EEG data. This rate was chosen to capture the fine-grained electrical activity of the brain accurately.
- **Connectivity:** Data was transmitted using the extender in pass-through mode. This setup involved connecting the headset to the extender, which was then connected to the PC via USB. The extender increased the battery life of the headset, allowing for continuous recording during longer sessions without the need for frequent recharging. This method ensured minimal movement constraints and facilitated a comfortable environment for the participant during the data collection sessions.
- **Sensor Material:** The headset uses saline-soaked felt pads for sensors, ensuring good conductivity and reducing skin impedance. This material choice helps in obtaining cleaner EEG signals with minimal artifacts.
- **Internal Filtering:** The EMOTIV EPOC X has built-in digital notch filters at 50Hz and 60Hz to eliminate power line interference, which is critical for maintaining data quality during recording.

3.2.4 Data Processing

The collected EEG data underwent several processing steps to prepare it for analysis in the EmotionWave dataset:

1. Preprocessing:

- **Downsampling:** The data were downsampled from the original 256 Hz to 200 Hz to align with the preprocessing protocols used in the SEED datasets.
- **Bandpass Filtering:** A bandpass filter ranging from 0.5 Hz to 75 Hz was applied to remove noise and artifacts from the EEG signals. Note that the EMOTIV EPOC X headset has a bandwidth range of 0.16 – 43 Hz with digital notch filters at 50Hz and 60Hz, which aids in reducing noise but does not replace the need for a broader bandpass filter for the purposes of this study.

2. Segmentation and Feature Processing:

- **Segmentation:** The preprocessed EEG data were segmented into windows based on a 1-second window size.

- **Labeling:** Each segment was labeled according to the emotion intended to be induced by the corresponding film clip. This labeling facilitated the creation of a structured dataset ready for further analysis and model training.
- **Feature Extraction:** Additional features such as differential entropy (DE), differential asymmetry (DASM), and rational asymmetry (RASM) were not created because they are not utilized in the emotion recognition model. The focus remained solely on the preprocessed, segmented EEG signals.

3. Data Storage:

- The segmented and labeled EEG data were stored in a structured format, with separate files for each session. Each file contained the segmented EEG data along with corresponding labels indicating the emotion categories: positive, negative, or neutral.

This detailed and systematic approach ensured the collection of high-quality EEG data, which formed the basis for subsequent emotion recognition model training and evaluation.

3.2.5 Implementation Details

This section provides a detailed overview of the implementation of the EEG emotion recognition experiment. The process is divided into several key stages: video preparation and experiment execution using the EmotivPRO Builder. Each stage is critical for ensuring the smooth operation and accurate data collection during the experiment.

3.2.5.1 Video Preparation

The video preparation stage involves two main tasks: concatenation of individual video clips and segmentation of the concatenated videos into 4-minute parts.

1. **Concatenation of Videos:** To begin with, video clips intended to elicit specific emotions (positive, neutral, and negative) were collected and stored in separate directories. Each directory corresponds to a specific emotion. The videos were concatenated into a single video file per emotion to facilitate easier playback during the experiment. This was achieved using a video processing library which concatenates multiple video files into a single continuous video. The output video is saved in a predetermined format (MP4), ensuring compatibility with the subsequent processing steps.
2. **Segmentation into 4-Minute Parts:** Once the videos were concatenated, they were further segmented into 4-minute parts. This segmentation is crucial for maintaining the structure of the experiment and ensuring that each trial is of manageable length. The segmentation process involves dividing the concatenated video into equal parts, each approximately 4 minutes long. If a segment is significantly shorter than the desired duration (less than 90% of the target length), it is discarded to maintain consistency across the dataset. Both the video and corresponding audio are segmented and saved separately, ensuring they can be played back in sync during the experiment.

3.2.5.2 Experiment Execution Using EmotivPRO Builder

Initially, an OpenCV-based program was developed to handle the experiment execution. This program included functionalities such as displaying video stimuli, collecting participant responses, and managing the flow of the experiment. However, due to the requirement for Cortex SDK access, this program was ultimately not used. Instead, the EmotivPRO Builder platform was utilized.

1. **Using EmotivPRO Builder:** The EmotivPRO Builder is an online platform designed for creating neuroscience experiments with Emotiv's EEG headsets. It allows for easy construction of experiments through various phases such as calibration, instruction, and stimuli presentation.
2. **Experiment Design:** Four separate experiments were created using the EmotivPRO Builder, each lasting approximately 55 minutes to constitute one session of EEG recording. Each experiment began with instructions for the participant, followed by headset calibration (15 seconds with eyes open looking at one spot, 15 seconds with eyes closed). Each trial within the experiment included the phases defined in Section 3.2.2.
3. **Event Marking:** The EmotivPRO Builder automatically marked the EEG recordings with events such as the start and stop of each video. The order of the emotions presented was positive, negative, neutral, and then repeating in this sequence.

This structured approach ensured the collection of high-quality EEG data, which formed the basis for subsequent emotion recognition model training and evaluation.

3.2.5.3 Functionalities of the OpenCV Program (Not Used)

The OpenCV program, although not used, was designed with several functionalities to enhance user experience and ensure accurate data collection. These details are included here for completeness:

1. **Experiment Protocol:** The experiment followed a structured protocol for each trial, the same one defined in Section 3.2.2. The sequence is designed to maximize participant engagement and ensure accurate emotion induction and recording.
2. **Displaying Video and Messages:** OpenCV was used to display video stimuli and messages to the participant. The video playback was handled within a full-screen window to minimize distractions and immerse the participant in the video content. Messages, such as hints about the upcoming trial or instructions for self-scoring, were displayed using OpenCV's text rendering capabilities. The text was centrally aligned and color-coded based on the type of message (e.g., hints, pauses).
3. **User Interaction:** Participants interacted with the system using keyboard inputs. Key functionalities included pausing the experiment (space bar), quitting (Q key), and skipping to the next screen (S key). When the experiment was paused, the screen displayed a paused message, and the system waited for the participant to resume. The

synchronization between the video and audio was maintained throughout the experiment to ensure accurate timing.

4. **Self-Scoring:** After each video clip, participants were prompted to score their emotional response. The scoring interface was displayed on the screen, and participants used the keyboard to input their scores. The scores were saved in a structured format for later analysis.
5. **Audio Synchronization:** The audio synchronization was managed by a custom audio player class, ensuring that the audio and video remained in sync throughout the experiment.
 - **Audio Player Implementation:** The audio player was implemented using the PyAudio library, which provided low-level audio playback functionalities. It opened the audio file, read the audio frames, and streamed them to the output device. The audio player class supported basic operations such as play, pause, resume, and seek, allowing for precise control over audio playback.
 - **Synchronization with Video:** During video playback, the system continuously monitored the synchronization between the audio and video streams. If a desynchronization was detected (e.g., audio and video were out of sync by more than 250 milliseconds), the audio player re-synced the audio to the current video position. This ensured that the participant's experience was not disrupted by timing issues. In practice, while experimenting, the audio was out of sync for at most 10 milliseconds even without this synchronization mechanism.
 - **Handling Pauses and Skips:** When the experiment was paused, the audio playback was also paused to maintain synchronization. Upon resuming, both the audio and video continued from the same point, ensuring continuity. If the participant chose to skip a video, the system stopped the current playback and proceeded to the next segment, updating the internal state accordingly.
6. **Functionalities of the Experiment:** The experiment included several functionalities to enhance the user experience and ensure accurate data collection.
 - **Pausing the Experiment:** Participants could pause the experiment at any time by pressing the space bar. During the pause, the screen displayed a paused message, and both video and audio playback were halted. This functionality allowed participants to take breaks without affecting the experiment's flow. It is important to note that this feature should generally not be used during the video playback sections to maintain the integrity of the emotional response being recorded.
 - **Skipping Segments:** Participants could skip to the next video segment by pressing the S key. This was useful if a participant felt that the current segment was not suitable or if there was an issue with the video playback, or they just didn't want to wait anymore.

- **Automatic Rest Periods:** After each self-scoring phase, a rest period was included to allow participants to relax and prepare for the next trial. This rest period was crucial for maintaining participant engagement and preventing fatigue. The rest duration was fixed, and the system automatically transitioned to the next trial after the rest period ended.

While these functionalities were developed, they were ultimately not employed due to the transition to using EmotivPRO Builder for the experiment execution.

3.2.5.4 Custom Dataset Loading

The loading of the EmotionWave dataset was implemented using the **EpocDataset-Loader** class. The dataset loading process included the following key steps:

- **EEG Data Loading:** EEG data were loaded from CSV files, ensuring that all necessary channels were included.
- **Bandpass Filtering:** A Butterworth bandpass filter was applied to filter frequencies between 0.5 Hz and 75 Hz.
- **Downsampling:** The EEG data were downsampled from 256 Hz to 200 Hz using polyphase re-sampling.
- **Segmentation:** The EEG data were segmented into 30-second windows for GAN-based augmentation and 1-second windows for the pre-training phase.
- **Labeling:** Each segment was labeled based on the emotional response recorded during the experiment.

3.3 Data Preprocessing

3.3.1 Data Loading

The SEED dataset is utilized to train the model. The loading process includes the following steps:

- **File Processing:** Each file is parsed to extract subject number, trial number, and EEG data.
- **Segmentation:** EEG data is segmented into windows based on a window size of 1 second.
- **Label Assignment:** Labels are mapped to each EEG segment based on trial numbers.

3.3.2 Data Augmentation

Data augmentation is applied to enhance the diversity of the training dataset and improve model robustness. The GAN was specifically used on the EmotionWave dataset. It was trained on segments of 15 seconds, which with 200 Hz resulted in training the GAN to create data of shape 1×3000 . A separate GAN was trained for each of the 14 channels of the EmotionWave dataset, which were concatenated into 14×3000 data. Overall, from the original dataset's size of 5760 EEGs of 1 second with dimensions 14×200 , after the GAN augmentation, we now have 11520 EEGs of with the same dimensions.

In addition to GAN-based augmentation, specific augmentations are applied in the time and frequency domains to simulate variability and improve the model's robustness.

GAN-based Augmentation

The paper [6] utilized Generative Adversarial Networks (GANs) to generate synthetic EEG data that accurately mimics the distribution of real EEG signals. This technique enhances data diversity and contributes to the training of more robust models.

In this study, the GAN model addresses data scarcity and class imbalance by generating additional EEG segments that exhibit statistical properties similar to the real data. These synthetic segments are then incorporated into the training set, thereby increasing its size and diversity. As a result, the augmented dataset improves the generalization and robustness of the emotion recognition model.

Time and Frequency Domain Augmentation

In addition to GAN-based augmentation, specific augmentations are applied in the time and frequency domains to simulate variability and improve the model's robustness.

1. Time Domain Augmentation:

- **Gaussian Noise Addition:** Gaussian noise with a specified standard deviation is added to the EEG signals to simulate variability.

2. Frequency Domain Augmentation:

- **Fourier Transformation:** EEG signals are converted from the time domain to the frequency domain using the Fast Fourier Transform (FFT).
- **Spectral Perturbation:** Frequency components are selectively perturbed by:
 - ◆ Zeroing out amplitudes that are below a threshold.
 - ◆ Scaling the others based on a uniform probability matrix and a specified perturbation range.

These time and frequency domain augmentations enhance model robustness by simulating real-world EEG variabilities. Combined with GAN-based augmentation, they significantly improve the training dataset's diversity and quality, leading to better emotion recognition performance.

3.4 Model Architecture

3.4.1 Overview

The proposed model architecture for EEG-based emotion recognition is designed to extract robust and generalizable features from EEG signals by leveraging both time and frequency domain information and aligning them in a shared latent space. This approach aims to improve the model's ability to recognize emotions accurately by utilizing the rich information contained in both domains.

The key components of this architecture include contrastive encoders, data augmentation techniques, cross-space projectors, and customized loss functions. The contrastive encoders are separate for the time and frequency domains, each tasked with extracting meaningful embeddings from the EEG signals. Data augmentation techniques are employed to create diverse training samples, thereby enhancing the model's robustness. Cross-space projectors are utilized to map the domain-specific embeddings to a shared latent space, ensuring consistency and alignment between the time and frequency domains. Customized loss functions are designed for each domain and an alignment loss is introduced to maintain consistency across these domains.

3.4.2 Pre-training

The pre-training phase is focused on learning generalizable representations from the EEG data without using labeled information. This phase initializes the model with features that can effectively capture the underlying patterns in EEG signals.

Data augmentation in this phase involves two main techniques. In the time domain, Gaussian noise is added to the original time-based signals to generate weakly augmented samples. This augmentation helps the model learn to recognize the core features of the signals despite slight variations. In the frequency domain, the original time-based signals are transformed into the frequency spectrum using Fourier Transformation. Perturbations are then introduced by adding or removing frequency components based on a uniform probability matrix, enhancing the model's ability to understand frequency-based variations.

The contrastive encoders consist of two components: the time-based encoder (ET) and the frequency-based encoder (EF). Both are designed as 2-layer transformers. The time-based encoder processes the time-domain signals to produce embeddings, while the frequency-based encoder processes the frequency-domain signals to generate corresponding embeddings.

Cross-space projectors are then used to map these embeddings into a shared latent space. The projector PT maps time-based embeddings, and the projector PF maps frequency-based embeddings to this space. This alignment is crucial for ensuring that the features extracted from both domains are comparable and can be used jointly.

As for loss functions, the time domain loss (LT) and frequency domain loss (LF) are using an NT-Xent loss that measures the similarity between augmented and original time- and frequency-based signals, respectively. Additionally, an alignment loss (LA) is employed to

align the time and frequency embeddings within the shared latent space, ensuring coherence between the two domains.

3.4.3 Fine-tuning

During the fine-tuning phase, the model is trained with labeled EEG data to optimize its performance for emotion recognition. This phase also incorporates the spatial relationships between EEG channels, leveraging a Graph Convolutional Network (GCN) to enhance spatial feature extraction.

The GCN construction involves creating an adjacency matrix using cosine similarity between EEG channels. This matrix facilitates the capture of spatial features by representing the relationships between different EEG channels. The graph convolution process employs K-order Chebyshev graph convolution to extract these spatial features more effectively, improving the model's understanding of the spatial dynamics within the EEG data.

The emotion classifier in this architecture is a 3-layer fully connected network. It comprises hidden layers with dimensions of 1024 and 128, followed by an output layer corresponding to the number of emotion classes. The classifier's objective is to achieve accurate emotion classification by leveraging the enriched embeddings from both the time and frequency domains.

Loss functions used in the fine-tuning phase include the cross-space losses (LT, LF, LA), which are NT-Xent losses applied similarly as in the pre-training phase to ensure continuity in learning. Additionally, a classification loss, specifically the cross-entropy loss, is used for training the emotion classifier to minimize the error in predicting the correct emotion classes.

3.4.4 GAN Architecture

The GAN models used in this study consist of a Generator and a Discriminator, each designed with specific layers to effectively model the EEG data. The Generator is responsible for producing synthetic EEG signals from random noise, while the Discriminator evaluates the authenticity of the generated signals.

3.4.4.1 Generator Architecture

The Generator takes a random noise tensor of size (batch_size, 1, latent_dim) as input and processes it through several layers to generate synthetic EEG data. The architecture can be seen in Table 3.1:

The first and last convolutional layers also include MaxPool1D and Dropout layers, while all convolutional layers are followed by **BatchNorm1D** and **LeakyReLU** activation. After the LSTM layer, we take only the last element in the channel dimension, finally passing through a Linear(128, 6000) layer.

Layer	Details	Output Size
Upsample	scale_factor = 30	$Batch_size \times 1 \times 3000$
Conv1D	(1, 128, kernel_size=50, stride=6, padding=22)	$Batch_size \times 128 \times 500$
Conv1D	(128, 128, kernel_size=8, stride=1, padding=4)	$Batch_size \times 128 \times 64$
Conv1D	(128, 128, kernel_size=8, stride=1, padding=3)	$Batch_size \times 128 \times 63$
Conv1D	(128, 128, kernel_size=8, stride=1, padding=3)	$Batch_size \times 128 \times 62$
LSTM	(input_size=16, hidden_size=128)	$Batch_size \times 128 \times 128$

Table 3.1: Generator Architecture.

3.4.4.2 Discriminator Architecture

The Discriminator architecture is similar to the Generator, but without the random tensor input and up-scaling layer. The Discriminator processes real and synthetic EEG data to classify them as authentic or generated.

3.4.4.3 Rationale for Design Choices

The design of the GAN models incorporates convolutional layers to extract frequency signals and an LSTM layer to learn temporal information. This combination allows the model to effectively capture the complex spatiotemporal patterns in EEG data. More specifically:

- **MaxAbsScaler:** MaxAbsScaler was used to normalize the EEG data because it scales each feature by its maximum absolute value, preserving the sparsity of the data and ensuring that the GAN operates within a stable range. This normalization is particularly important for EEG data due to its wide dynamic range.
- **LSTM Layer:** The LSTM layer is critical for capturing the temporal dependencies in the EEG signals. By learning the temporal patterns, the GAN can generate more realistic sequences that reflect the inherent temporal structure of the EEG data.

3.4.4.4 Training and Optimization

The GAN is trained using Binary Cross-Entropy Loss (BCELoss). The optimizers used for both the Generator and Discriminator are Adam Optimizers with learning rates of $2e-4$ and betas (0.5, 0.999). Learning rate scheduler, ReduceLROnPlateau with a patience of 10 epochs, is employed for the Generator to adjust the learning rate dynamically, ensuring efficient training.

This detailed architecture and training strategy ensure that the GANs effectively generate high-quality synthetic EEG data, which is then used to augment the training dataset for the emotion recognition model.

3.5 Model Training

The training process involves two main phases: pre-training and fine-tuning. The model was trained on 70% of the original data, and in cases where data was generated with the

GAN, it was all added into the training data.

3.5.1 Pre-training Phase

In the pre-training phase, the model begins by preparing the dataset, where unlabeled EEG data is augmented and fed into the contrastive encoders. The loss calculation involves computing the time contrastive loss, frequency contrastive loss, and alignment loss using NT-Xent. These losses guide the model to learn representations that are robust and generalizable. The model parameters are then updated via backpropagation using the combined loss function, progressively improving the model's feature extraction capabilities.

3.5.2 Fine-tuning Phase

The fine-tuning phase starts similarly with data preparation, where labeled EEG data is augmented and processed through the pre-trained encoders. It additionally passes through a K-order Chebyshev graph convolution to extract spatial features using the constructed adjacency matrix. The emotion classifier is then trained with cross-entropy loss to fine-tune the model for accurate emotion recognition. The overall fine-tuning loss, which includes the NT-Xent losses and cross-entropy loss, is used to update the model parameters. This comprehensive training approach ensures that the model is well-optimized for recognizing emotions in EEG data.

3.6 Model Evaluation

The model evaluation involved calculating the evaluation loss using the NT-Xent loss on 30% of the original data. It is important to note that the evaluation was performed on the original augmented data without the GAN data.

3.6.1 Evaluation Metrics

The evaluation metrics included a simple accuracy calculation. The accuracy was determined by checking each prediction against the true labels, summing the correct predictions, and dividing by the total number of samples.

3.6.2 Experimental Setup

The experimental setup involved using the pre-trained model, fine-tuning it with the labeled data, and evaluating it on a separate dataset. The hardware and software configurations included:

- **Hardware:** NVIDIA GeForce RTX 4090 TI GPU.
- **Software:** Python 3.12, PyTorch 2.3.0.

The experiments were conducted over multiple epochs, with the best-performing model based on validation accuracy being selected for final evaluation.

Chapter 4

Results and Discussion

4.1 Hyper-parameter Tuning Experiments

4.1.1 Optimizing DataLoader Parameters

Overview

The goal of this experiment was to optimize the **num_workers** and **prefetch_factor** parameters of the **DataLoader** to minimize the data loading time and reduce GPU idle time. These tests were essential to ensure that the data loading process was efficient and did not become a bottleneck during model training.

Experimental Setup

The experiments tested various combinations of **num_workers** and **prefetch_factor** values over 50 epochs. The performance metric used was the overall elapsed time for each combination, with the aim of identifying the configuration that resulted in the shortest time.

Results

The results of the experiments are summarized in the Table 4.1.

Discussion

From Table 4.1, we conclude that the optimal configuration was **num_workers = 5** and **prefetch_factor = 2**, achieving the lowest elapsed time of 32 minutes and 56 seconds per epoch. The rationale for these tests was to minimize the time the GPU was idle due to waiting for data, thereby improving overall training efficiency. This configuration provided a balance between sufficient parallel data loading processes and an optimal amount of data prefetched, reducing bottlenecks in data transfer.

Num Workers	Prefetch Factor	Time (HH:MM:SS.milliseconds)
5	2	0:32:56.436
6	1	0:33:04.665
5	4	0:33:11.745
3	2	0:33:20.534
8	1	0:33:29.338
4	2	0:33:30.620
4	1	0:33:32.329
8	2	0:33:36.731
4	4	0:33:36.828
7	4	0:33:40.763
7	2	0:33:43.519
7	1	0:33:45.172
3	1	0:33:45.564
6	2	0:33:53.110
5	1	0:33:56.519
2	4	0:34:02.076
6	4	0:34:04.084
3	4	0:34:06.625
8	4	0:34:11.154
2	2	0:34:16.823
2	1	0:36:11.101

Table 4.1: Results of DataLoader Parameter Optimization.

4.1.2 Optimizing Pre-training Parameters

Overview

This experiment aimed to determine what the best values of **scheduler_patience** and **early_stopping_patience** are for the **PreTraining** class. These parameters are crucial for the adaptive learning rate scheduler and early stopping mechanism, which help prevent overfitting and ensure efficient training.

Experimental Setup

The experiments tested different combinations of the values of **scheduler_patience** and **early_stopping_patience** over 100 epochs. The performance metric was the final training loss, with the goal of finding the combination that yielded the lowest loss. The results of the experiments are summarized in the Table 4.2.

Discussion

From Table 4.2, we conclude that the best results were obtained with **scheduler_patience = 50** and **early_stopping_patience = 10**, resulting in the lowest final loss of -2.6390. These tests were conducted to optimize the learning rate adjustment and early stopping criteria, ensuring the model could efficiently converge without overfitting. By identifying the optimal

Scheduler Patience	Early Stopping Patience	Final Loss
50	10	-2.6390
25	10	-2.5647
10	25	-2.4591
50	25	-2.4308
25	25	-2.4073
10	10	-2.3840

Table 4.2: Results of Pre-training Parameter Optimization.

patience values, the training process becomes more stable and less likely to terminate prematurely, thereby improving the overall performance of the pre-training phase.

4.1.3 Conclusion

These experiments were crucial for fine-tuning the training process of the EEG-based emotion recognition model. By optimizing data loading parameters, the efficiency of GPU utilization was significantly improved. Additionally, refining the patience parameters for the scheduler and early stopping mechanism ensured that the model training was both efficient and effective, leading to better performance and generalization capabilities.

4.2 Model Experimental Results

4.2.1 Pre-training with SEED Dataset

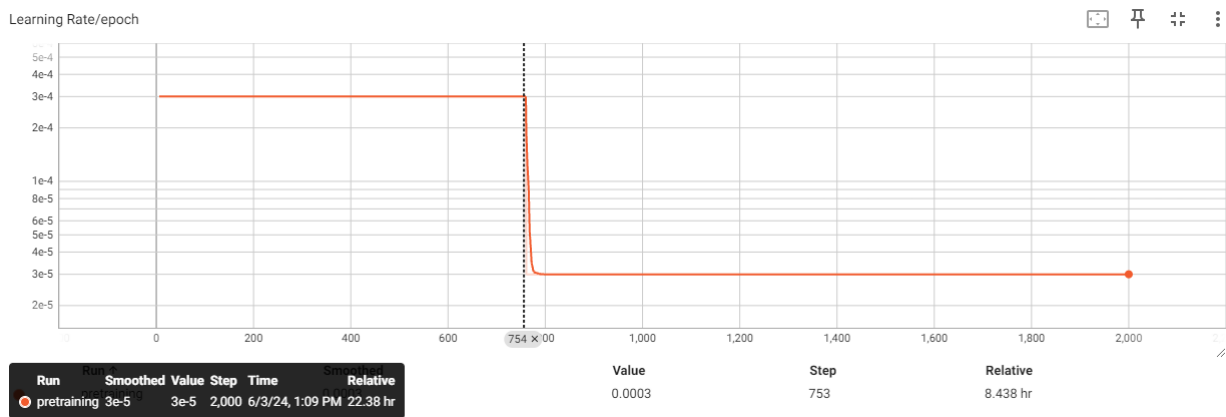


Figure 4.1: SEED pre-training learning rate plot.

The pre-training phase of our model using the SEED dataset involved monitoring the learning rate and loss over epochs to evaluate the performance and effectiveness of our training strategy. The learning rate scheduler utilized was **ReduceLROnPlateau**, which dynamically reduces the learning rate when the monitored metric (in this case, the loss) stops improving. The patience parameter for the scheduler was set to 50, meaning the learning rate was reduced after 50 epochs without significant improvement in the loss.

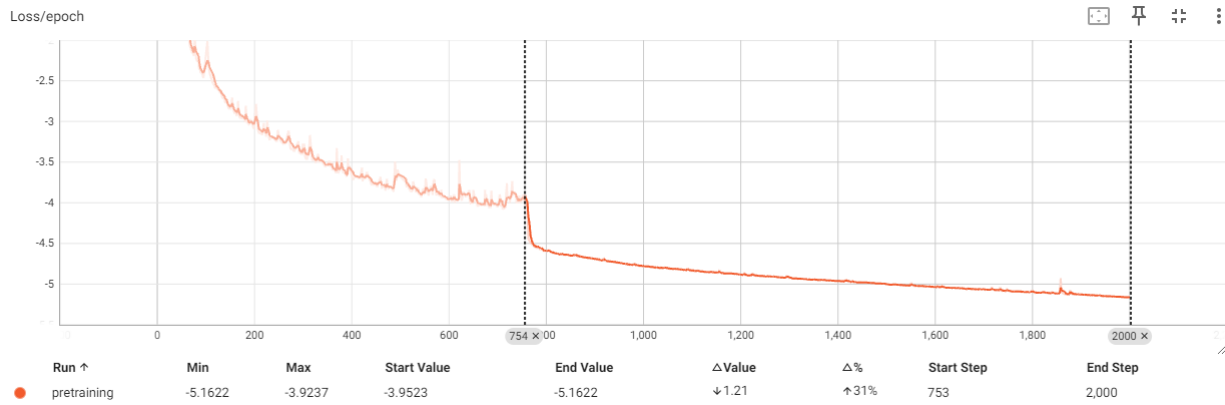


Figure 4.2: SEED pre-training loss plot.

As observed in Figure 4.1, the learning rate remained steady at $4e-4$ for the initial 753 epochs. However, at epoch 754, due to the lack of improvement in the loss, the scheduler reduced the learning rate to $3e-5$. This reduction in learning rate allowed the model to fine-tune the weights more precisely, leading to a noticeable improvement in the loss.

Correspondingly, Figure 4.2 illustrates the change in the loss over epochs. Initially, the loss decreases gradually, showing a general trend of improvement. However, it is evident that around epoch 754, there is a marked improvement in the loss reduction rate, coinciding with the reduction in the learning rate. This improvement suggests that the model benefited from a lower learning rate, enabling it to escape potential local minima and continue to optimize effectively.

The sharp decrease in loss post-epoch 754 highlights the importance of adaptive learning rate schedules in training deep learning models. By reducing the learning rate, the model can achieve a more stable convergence, as seen in the sustained decrease in loss following the adjustment. This adaptive approach ensures that the model does not overfit early on and continues to learn from the data throughout the training process.

These results demonstrate the effectiveness of the **ReduceLROnPlateau** scheduler in enhancing model performance during pre-training. The careful adjustment of learning rates in response to the monitored loss metric significantly contributes to the model's ability to learn robust and generalizable features from the EEG data. The pre-training process took approximately 22 hours to complete, indicating the computational effort and time investment required to achieve these results.

4.2.2 Fine-tuning with SEED Dataset

The fine-tuning phase of our model was conducted using the SEED dataset, with the evaluation being carried out on 30% of the dataset, stratified for the verdicts of the EEG data. The overall fine-tuning process took approximately 9 hours.

Figure 4.3 depicts the training loss per epoch. The training loss decreases consistently throughout the training process, reaching a minimum of 2.7832 at epoch 20. The continuous decrease in training loss indicates effective learning and optimization of the model parameters during fine-tuning.

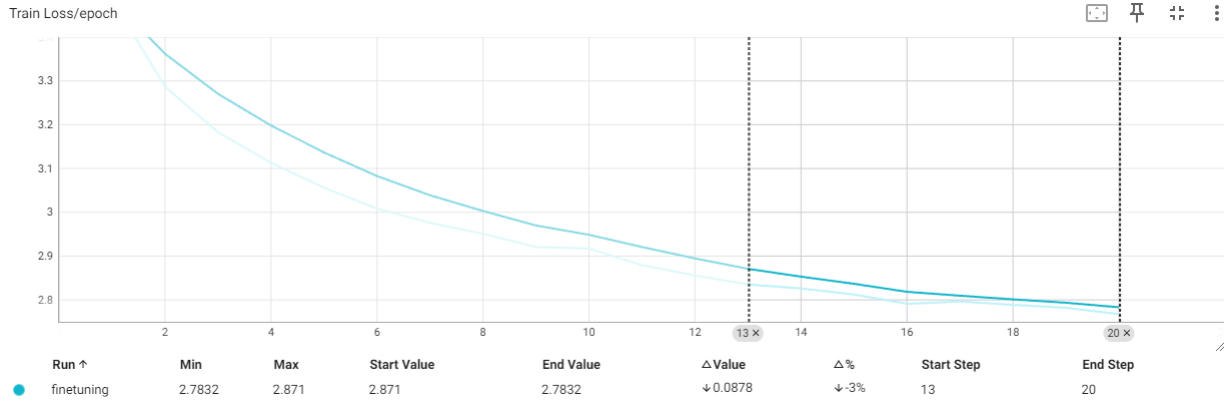


Figure 4.3: SEED fine-tuning training loss plot.

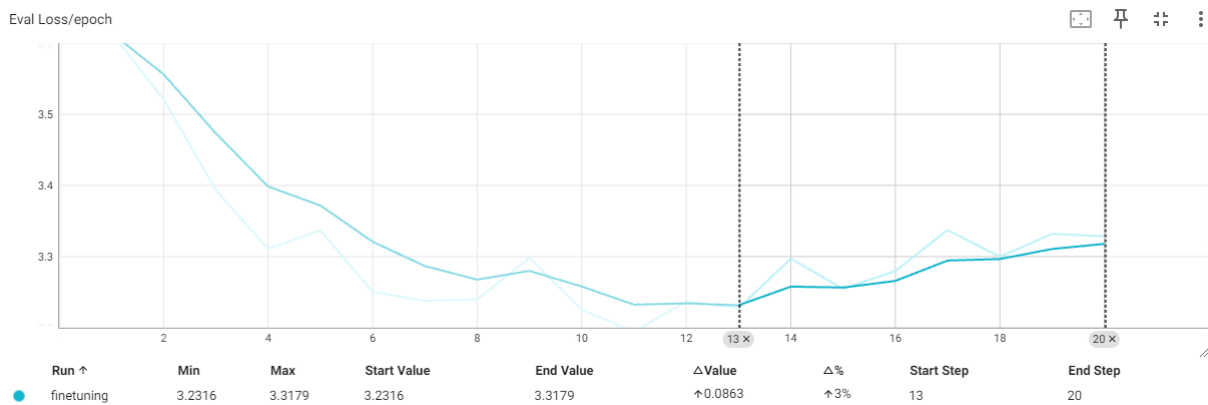


Figure 4.4: SEED fine-tuning evaluation loss plot.

Figure 4.4 shows the evaluation loss per epoch. Initially, the evaluation loss decreases significantly, reaching a minimum value of 3.2316 at epoch 13. However, after epoch 13, the evaluation loss starts to show slight fluctuations, potentially indicating the model's performance on the evaluation dataset has started to stabilize. The final evaluation loss at epoch 20 is 3.3179, representing an overall increase of 0.0863 from the minimum value observed. To avoid overfitting, the training was halted at this point.

Figure 4.5 illustrates the evaluation accuracy per epoch. The evaluation accuracy shows a consistent upward trend, starting from 50.78% and peaking at around 74% starting with epoch 13. Although there are slight fluctuations in accuracy after epoch 13, the overall trend indicates that the model maintains a high level of accuracy on the evaluation dataset. The highest accuracy of 74.13% reflects an overall improvement of 23.35% from the initial accuracy, which was derived from the pre-trained model.

The steady decrease in both training and evaluation losses, combined with the increase in evaluation accuracy, demonstrates the effectiveness of the fine-tuning process. The fluctuations in evaluation loss and accuracy post-epoch 13 could be attributed to the model reaching its capacity to generalize on the validation set, indicating a balance between fitting the training data and maintaining performance on the evaluation set.

These results underscore the importance of fine-tuning in enhancing the model's performance, particularly in adapting pre-trained models to specific datasets. The stratified

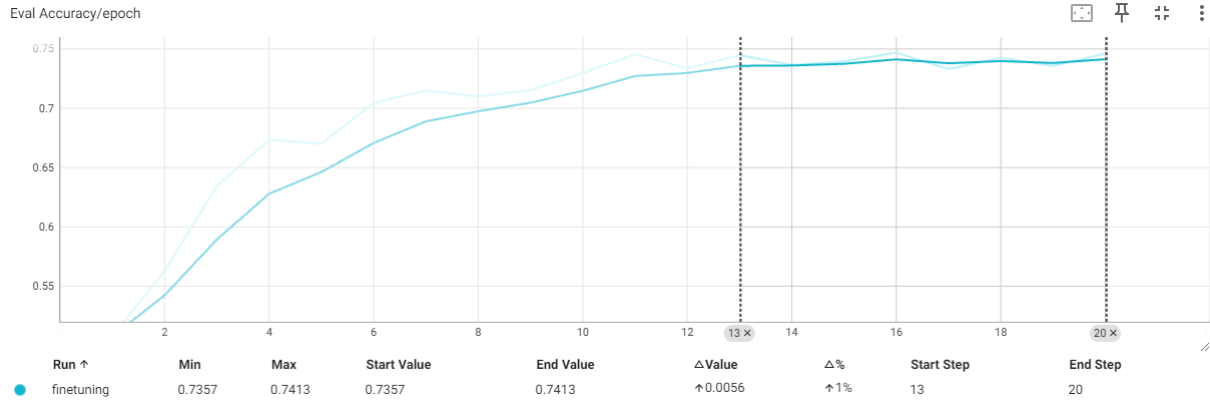


Figure 4.5: SEED fine-tuning evaluation accuracy plot.

evaluation ensures that the model's performance metrics are reliable and representative of its generalization capability across different classes in the EEG data.

4.2.3 Pre-training with our EmotionWave Dataset

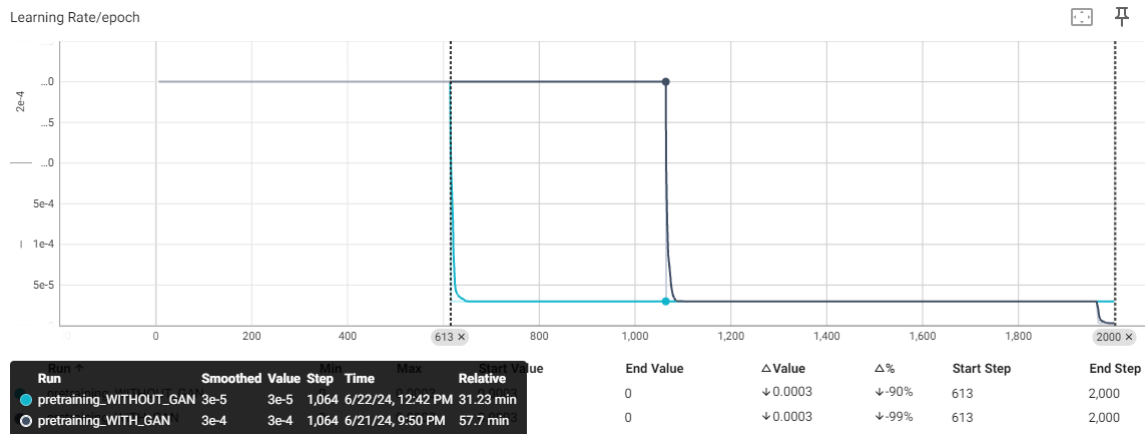


Figure 4.6: EmotionWave pre-training learning rate plot.

The pre-training phase of our model using the EmotionWave dataset was conducted with two configurations: with and without GAN augmentation. As shown in Figure 4.6, the learning rate for the non-GAN model decreased at epoch 613, while for the GAN-augmented model, it decreased later at epoch 1064. This earlier adjustment for the non-GAN model indicates faster convergence, allowing more precise weight adjustments and stable learning dynamics. The learning rate reduction facilitated continued optimization and prevented overfitting in the non-GAN model.

Figure 4.7 illustrates the loss over epochs for both configurations. The non-GAN model demonstrates a consistent decrease in loss post-epoch 613, reflecting efficient learning and optimization. In contrast, the GAN-augmented model, despite a later learning rate reduction, shows a slower loss reduction rate. This suggests that while GAN augmentation introduces beneficial variability for generalization, it also adds complexity, making it harder to optimize

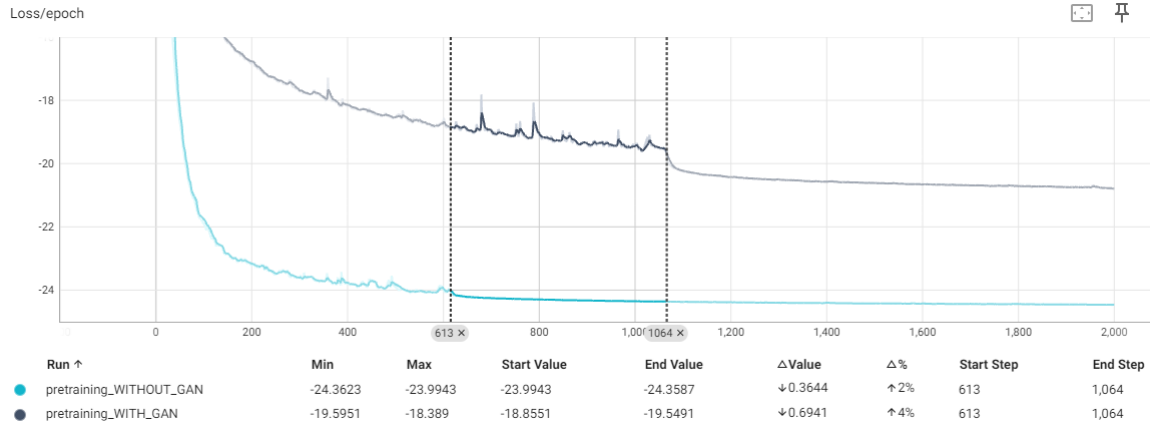


Figure 4.7: EmotionWave pre-training loss plot.

effectively. These results highlight the importance of balancing augmentation complexity with adaptive learning rate schedules to achieve optimal pre-training performance.

GAN

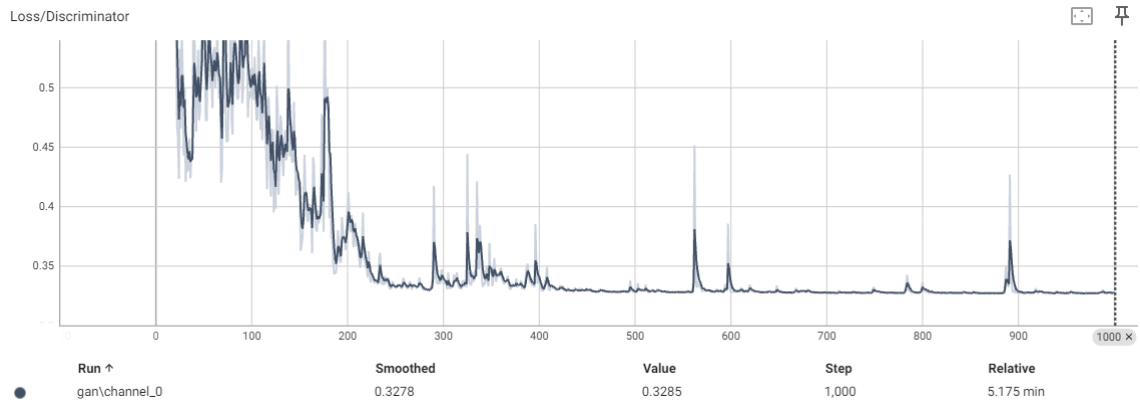


Figure 4.8: EmotionWave GAN Discriminator loss plot for channel 0 of EEGs.

The GAN training phase involved generating synthetic EEG data to augment the EmotionWave dataset. Figures 4.8, 4.9, and 4.10 provide insights into the training dynamics of the GAN components.

Figure 4.10 illustrates the learning rate progression for the generator. The learning rate was initialized at $2e-3$ and reduced after approximately 200 epochs. Despite this reduction, the generator's ability to optimize did not significantly improve, suggesting that the initial high learning rate might have led to sub-optimal weight adjustments that the later reductions could not fully correct.

Figure 4.8 shows the loss for the discriminator over epochs. The discriminator's loss decreases and stabilizes after about 250 epochs, indicating that it effectively learns to distinguish between real and synthetic EEG signals early in the training process. This stability suggests that the discriminator quickly adapts to its task and maintains a robust performance throughout the training.

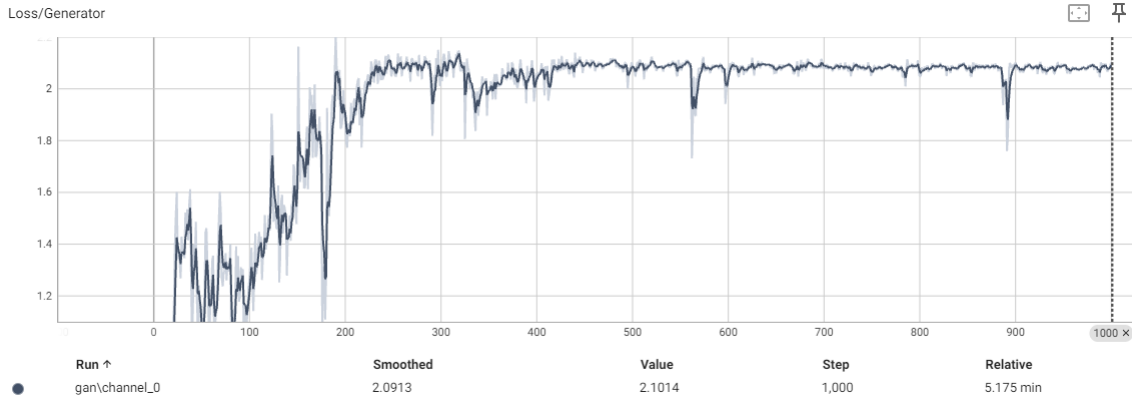


Figure 4.9: EmotionWave GAN Generator loss plot for channel 0 of EEGs.

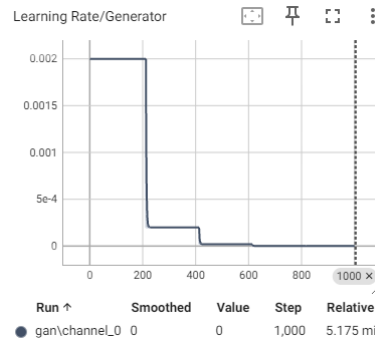


Figure 4.10: EmotionWave GAN Generator learning rate plot for channel 0 of EEGs.

Conversely, as seen in Figure 4.9, the generator's loss exhibits more variability and does not show the same level of stabilization. The generator's loss indicates that it struggles to produce high-quality synthetic data that can consistently deceive the discriminator. This is further evidenced by the loss plateauing after the initial learning rate reduction, implying that the generator does not gain additional improvements in generating realistic EEG data despite the adjusted learning rate.

These results highlight a disparity in the learning dynamics between the generator and discriminator. While the discriminator adapts quickly and maintains stability, the generator faces challenges in optimizing its performance. This indicates the need for a more balanced learning rate strategy or additional training techniques to enhance the generator's capability to produce high-quality synthetic EEG data.

4.2.4 Fine-tuning with our EmotionWave Dataset

Without GAN in Pre-training

The fine-tuning phase of the EmotionWave dataset, using models pre-trained without GAN augmentation, was assessed for performance. Figures 4.11, 4.12, and 4.13 depict the training loss, evaluation loss, and evaluation accuracy, respectively, for configurations with and without GAN during fine-tuning.

Figure 4.11 shows that the training loss for the non-GAN pre-trained model decreases

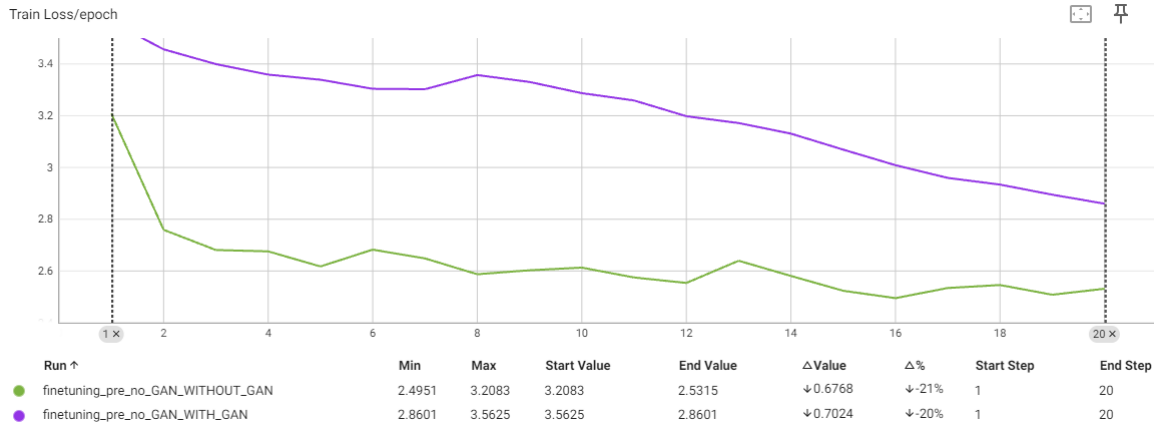


Figure 4.11: EmotionWave pre-training without GAN-generated data train loss plot.

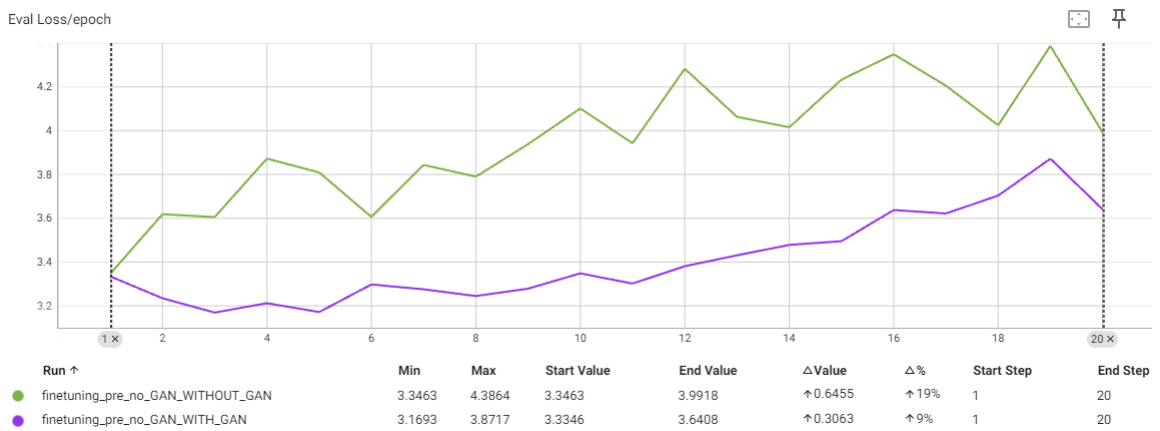


Figure 4.12: EmotionWave pre-training without GAN-generated data evaluation loss plot.

steadily, indicating effective optimization during fine-tuning. However, Figure 4.12 reveals that the evaluation loss for this configuration does not follow the same trend, instead exhibiting fluctuations and an overall increase. This divergence between training and evaluation loss suggests potential overfitting, where the model fits the training data well, but struggles to generalize to unseen data. The presence of GAN during fine-tuning appears to aid in reducing the evaluation loss initially, but this advantage diminishes over epochs, indicating the need for further refinement in training strategies or dataset quality.

Figure 4.13 illustrates the evaluation accuracy over epochs. The model fine-tuned with GAN augmentation reaches its peak accuracy of 66.45% at epoch 5, outperforming the configuration without GAN augmentation. This peak performance suggests that the GAN-augmented fine-tuning phase enhances the model's ability to generalize better to the evaluation dataset. However, the accuracy fluctuates beyond this point, indicating instability in learning, possibly due to the inherent variability and noise in the dataset.

With GAN in Pre-training

The fine-tuning phase of the EmotionWave dataset using models pre-trained with GAN augmentation was evaluated for performance. Figures 4.14, 4.15, and 4.16 illustrate the training loss, evaluation loss, and evaluation accuracy, respectively, for configurations with

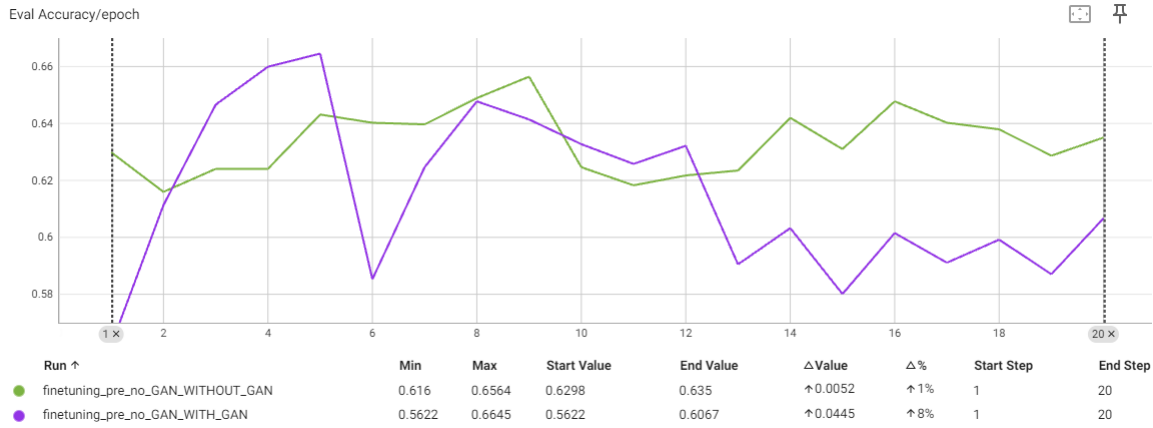


Figure 4.13: EmotionWave pre-training without GAN-generated data evaluation accuracy plot.

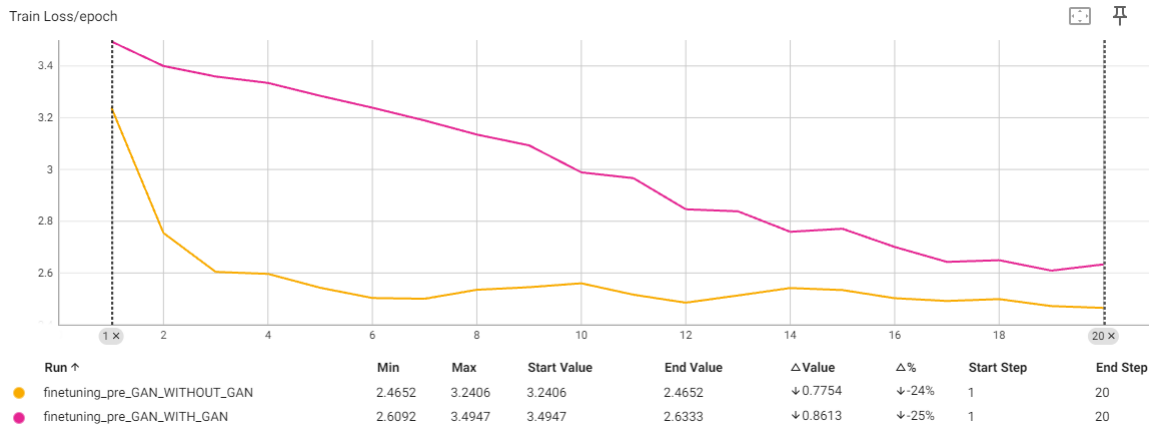


Figure 4.14: EmotionWave pre-training with GAN-generated data train loss plot.

and without GAN during fine-tuning.

Figure 4.14 shows that the training loss for the model fine-tuned without GAN decreases and stabilizes quickly, within about 6 epochs. Conversely, the training loss for the model fine-tuned with GAN continues to decrease gradually throughout the fine-tuning process. However, this continuous improvement in training loss does not translate to better evaluation performance, as seen in Figure 4.15. The evaluation loss for both configurations increases over time, similar to the trend observed in Section 4.2.4. This increase in evaluation loss suggests that while the models continue to optimize for the training data, they struggle to generalize to the evaluation dataset, possibly due to overfitting or inherent noise in the data.

Figure 4.16 presents the evaluation accuracy across epochs. The evaluation accuracy fluctuates significantly, further highlighting the dataset's variability and potential issues. The highest accuracy achieved with GAN pre-training is lower than that of the model without GAN pre-training, as noted in the previous section. The model fine-tuned without GAN shows initial improvement, but the gains are not sustained, indicating that the benefits of GAN pre-training may not be fully realized in this configuration.

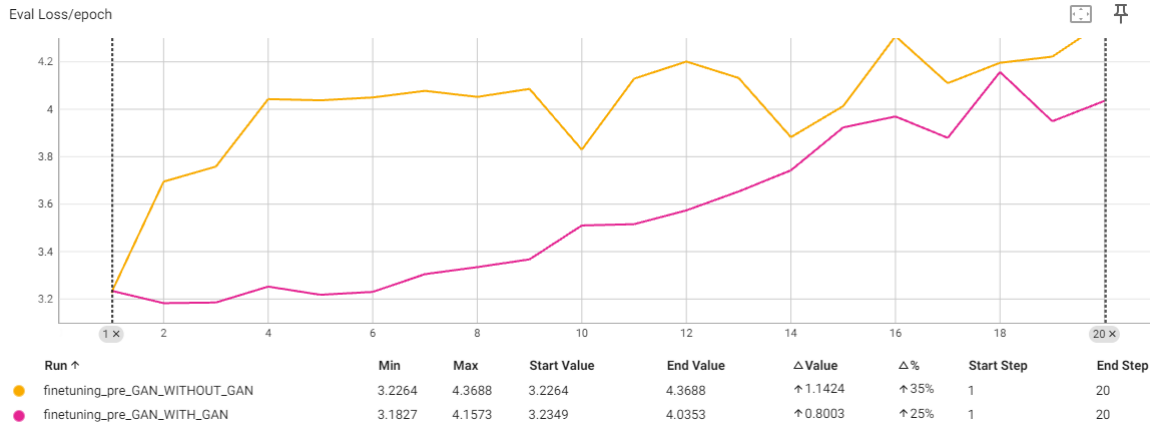


Figure 4.15: EmotionWave pre-training with GAN-generated data evaluation loss plot.

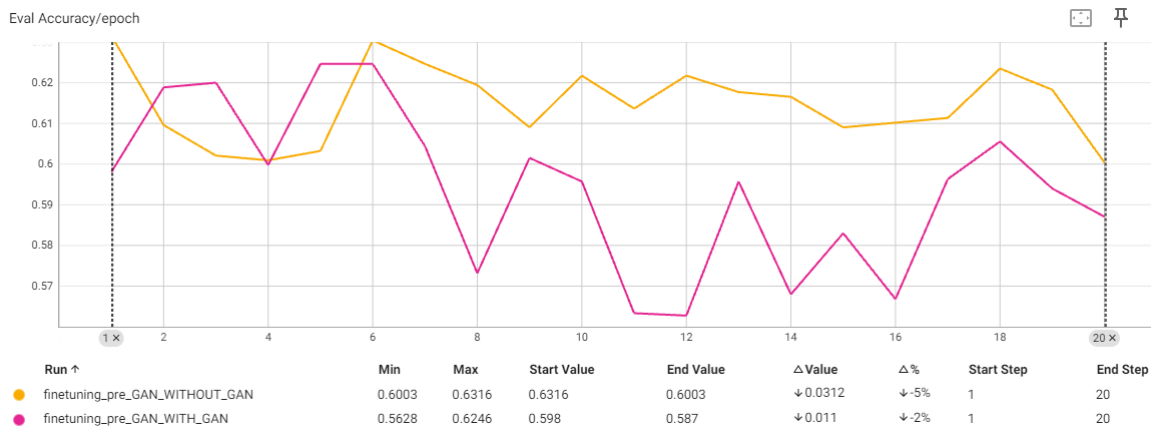


Figure 4.16: EmotionWave pre-training with GAN-generated data evaluation accuracy plot.

Conclusions

The inconsistent performance and fluctuations in evaluation metrics underscore potential issues with the dataset itself. The observed patterns suggest that the dataset may contain noise or outliers that impact model training and evaluation. Future work should consider manual filtering and cleaning of the dataset to enhance its quality and ensure more stable and reliable model performance. Additionally, balancing the complexity introduced by GAN-generated data with the learning dynamics of the model remains crucial for optimizing performance.

These observations emphasize the importance of dataset quality and the fine-tuning process in achieving robust and generalizable emotion recognition models. The results highlight the potential benefits and challenges of incorporating GANs in the training pipeline, providing valuable insights for further improvements.

4.3 Comparative Analysis

This section provides a detailed comparison of the performance of EEG-based emotion recognition models trained on the SEED dataset and the EmotionWave dataset. The

analysis focuses on evaluating both the pre-training and fine-tuning phases, with a specific emphasis on comparing models with and without GAN augmentation during pre-training.

4.3.1 Pre-training Phase Comparison

For the SEED dataset, the pre-training phase exhibited a consistent decrease in loss, demonstrating effective optimization facilitated by adaptive learning rate adjustments. The learning rate for the SEED dataset was reduced at epoch 754, which resulted in a significant improvement in loss reduction. This adaptive strategy proved to be effective, ensuring the model could continue optimizing without overfitting. The entire pre-training phase for the SEED dataset took approximately 22 hours to complete, reflecting the substantial computational effort required.

In contrast, the EmotionWave dataset showed different characteristics in its pre-training phase. When trained without GAN augmentation, the model demonstrated a steady decrease in loss with an earlier learning rate reduction at epoch 613, indicating faster convergence and more stable learning. However, when GAN augmentation was incorporated, the model exhibited a slower rate of loss reduction, with the learning rate adjustment occurring later at epoch 1064. This slower convergence suggests that the added complexity from GAN augmentation presented optimization challenges. The pre-training phase for the EmotionWave dataset took about 1 hour, primarily due to the smaller size of the dataset.

4.3.2 Fine-tuning Phase Comparison

During the fine-tuning phase, the SEED dataset continued to show robust performance. The training loss decreased continuously, reflecting effective model optimization. The evaluation loss stabilized with slight fluctuations, suggesting that the model generalized well to unseen data. The SEED model achieved an evaluation accuracy of 74.13%, indicating a substantial improvement from the pre-training phase and underscoring the model's capability to adapt and learn effectively from the data.

For the EmotionWave dataset, the fine-tuning phase presented more challenges. When the model was pre-trained without GAN augmentation, it exhibited a consistent decrease in training loss, but the evaluation loss showed fluctuations, indicating potential overfitting. Despite these fluctuations, the peak evaluation accuracy reached 66.45% at epoch 5 when fine-tuning with GAN data, demonstrating that GAN data provided some benefit during fine-tuning. However, when the model was pre-trained with GAN augmentation, the training loss improved continuously, but the evaluation loss again showed fluctuations. This configuration resulted in lower peak accuracy compared to the non-GAN configuration, suggesting that the complexities introduced by GAN data during pre-training were not fully resolved during fine-tuning.

4.3.3 Discussion

The comparative analysis reveals that the SEED dataset benefited from a more stable training process and achieved higher accuracy during fine-tuning compared to the EmotionWave dataset. The stability and higher accuracy of the SEED model during fine-tuning indicate that the dataset's quality and the pre-training strategies were more effective in ensuring robust generalization. In contrast, the EmotionWave dataset faced challenges with evaluation loss stability and consistency in peak accuracy. While the introduction of GAN augmentation increased data diversity and provided some benefits, it also added complexity to the training process, which affected the optimization dynamics.

These findings underscore the critical importance of dataset quality and the need for a balanced approach to data augmentation. Effective optimization requires not only robust data augmentation strategies, but also careful management of the complexities they introduce. Future work should focus on enhancing the preprocessing techniques to further reduce artifacts in the dataset and improving GAN training methods to ensure balanced progress between the generator and discriminator. Additionally, exploring more advanced data augmentation and preprocessing strategies could improve the overall quality and robustness of emotion recognition models.

Chapter 5

Conclusions

5.1 Summary of Findings

This research introduced the EmotionWave dataset, created using the EMOTIV EPOC X headset, to address the challenges in EEG-based emotion recognition. By leveraging the Joint Contrastive Learning framework with Feature Alignment (JCFA) and incorporating GAN-based data augmentation, the study aimed to enhance model performance. The comparative analysis between the SEED and EmotionWave datasets revealed that while pre-training with GAN-augmented data initially showed promise, the highest evaluation accuracy was achieved with a model pre-trained without GAN data, but fine-tuned with GAN data, achieving a peak accuracy of 66.45%. The SEED dataset consistently outperformed the EmotionWave dataset in terms of stability and peak accuracy during fine-tuning, underscoring the importance of dataset quality and optimization strategies.

5.2 Contributions

This thesis makes several contributions to the field of EEG-based emotion recognition:

- The development of the EmotionWave dataset, tailored specifically for emotion recognition with the EPOC X headset, providing a new resource for future research.
- The implementation of GANs for effective data augmentation, which increased the dataset's size and diversity, demonstrating the potential of GANs in addressing data scarcity and variability.
- The adaptation of the JCFA framework for the 14-channel EEG configuration of the EPOC X, showcasing the framework's flexibility and applicability to different EEG setups.
- Detailed experimentation and analysis, demonstrating the impact of GAN-augmented data on model performance and providing insights into the complexities introduced by such augmentation.

5.3 Limitations

Despite the advancements achieved, this study faced several limitations:

- **Dataset Quality:** The inherent difficulty in creating a high-quality dataset led to lower accuracy due to remaining artifacts in the data, despite extensive preprocessing efforts.
- **GAN Training Challenges:** The discriminator often trained faster than the generator, hindering the overall performance of the GAN and complicating the training process.
- **Overfitting Risks:** The potential for overfitting, especially during fine-tuning, affected the model's ability to generalize effectively, as indicated by fluctuations in evaluation loss and accuracy.

5.4 Future Work

Future research should build on these findings by addressing the following areas:

- **Improved Preprocessing:** Enhancing preprocessing techniques to further reduce artifacts and noise in the dataset, thereby improving the overall data quality.
- **GAN Training Enhancements:** Developing improved GAN training methods to ensure balanced progress between the generator and discriminator, thereby enhancing the quality of synthetic data.
- **Additional Data Augmentation:** Exploring more advanced data augmentation and preprocessing strategies to improve dataset quality and model robustness, potentially incorporating multimodal data sources.
- **Manual Data Cleaning:** Increasing the dataset quality by manually removing obvious artifacts in the EEG data and gathering more data to facilitate better model learning and generalization.
- **Balanced Complexity:** Finding an optimal balance between the complexity introduced by GAN-generated data and the model's learning dynamics to enhance overall performance, ensuring that augmentation strategies do not adversely impact optimization.

By addressing these areas, future research can contribute to the development of more robust and accurate EEG-based emotion recognition systems, potentially enhancing their applicability in real-world scenarios across various fields such as healthcare, entertainment, and human-computer interaction. These efforts will help to create more reliable and scalable emotion recognition technologies, further advancing the state of the art in EEG-based emotion recognition.

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